DECLARATION

I, FORLEMU NKENGLEFACK GUY STEPHANE declare that this piece of work titled "**DESIGN AND IMPLEMENTATION OF A VIRTUAL UNIVERSITY ECOSYSTEM FOR SCALABLE DIGITAL EDUCATION IN DEVELOPING COUNTRIES"** case of "**ESCHOSYS TECHNOLOGIES**" is research carried out by the researcher within a period of three months. This work is my personal effort and has never been presented or publish in any form and all borrowed ideas are duly acknowledged.

SIGNATURE: ____

DATE: ____

CERTIFICATION

This is to certify that this piece of work entitled the **“DESIGN AND IMPLEMENTATION OF A VIRTUAL UNIVERSITY ECOSYSTEM FOR SCALABLE DIGITAL EDUCATION IN DEVELOPING COUNTRIES**” case of “**ESCHOSYS TECHNOLOGIES**”, is an original work done by FORLEMU NKENGLEFACK GUY STEPHANE which was realized under my supervision and submitted in CITEC-HITM Yaoundé as a condition for partial fulfillment for the award of a Bachelor Degree in Computer Software Engineering.

SUPERVISOR: MR. ABANJI DIEUDONNE

SIGNATURE OF THE SUPERVISOR: ____

DATE: ____

DEDICATION

To the FORLEMU FAMILY

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ABSTRACT

This project presents an **Online University Management System** designed to create a digital academic **ecosystem** that enhances teaching, learning, and administration in **developing countries**. The platform integrates features for **virtual** lectures, course delivery, academic record management, and seamless communication among students, lecturers, and administrators. It supports secure authentication, assignment submissions, grading, and notifications while offering interactive dashboards for personalized access. Built with Node.js, Express, MongoDB, and Next.js, the system ensures scalability, cross-platform usability, and reliability. By streamlining academic processes, this project contributes to a more accessible and efficient **online university** framework.

Keywords: Online, University, Ecosystem, Virtual, Developing Countries

RESUME

Ce projet présente un **Système de Gestion d'Université en Ligne** conçu pour créer un **écosystème** académique numérique visant à améliorer l'enseignement, l'apprentissage et l'administration dans les **pays en développement**. La plateforme intègre des fonctionnalités pour les cours **virtuels**, la diffusion des enseignements, la gestion des dossiers académiques ainsi que la communication fluide entre étudiants, enseignants et administrateurs. Elle prend en charge l'authentification sécurisée, le dépôt des devoirs, la notation et les notifications, tout en offrant des tableaux de bord interactifs pour un accès personnalisé. Développé avec Node.js, Express, MongoDB et Next.js, le système garantit évolutivité, compatibilité multiplateforme et fiabilité. En rationalisant les processus académiques, ce projet contribue à un cadre d'**université en ligne** plus accessible et efficace.

Mots-clés : En ligne, Université, Écosystème, Virtuel, Pays en développement

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CHAPTER ONE

INTRODUCTION AND PRESENTATION OF ENTERPRISE

INTRODUCTION

In recent years, the advancement of digital technologies has significantly transformed various sectors of society, with education being one of the most impacted. The traditional model of education, which typically involves students attending physical classrooms and engaging in face-to-face interactions with instructors, has gradually been evolving to include digital alternatives. This transformation has been further accelerated by global events, such as the COVID-19 pandemic, which forced institutions worldwide to reconsider how education is delivered. In this context, the need for robust and reliable online learning platforms has become more pressing than ever before.

The idea of a Virtual University platform emerges as a direct response to this evolving educational landscape. Unlike typical e-learning systems that are often designed as supplementary tools for physical institutions, this project envisions a complete, fully virtual university experience one in which students can not only take courses online but also complete their entire application, admission, and academic journey without ever setting foot in a traditional campus. This approach addresses several persistent challenges in higher education, especially in regions where access to quality education is limited by geography, financial constraints, or inadequate infrastructure.

1.1. BACKGROUND OF THE STUDY

The background of the study includes; The **Historical background, Conceptual background, Theoretical background and Contextual background** of this Virtual University Platform.

1.1.1. HISTORICAL BACKGROUND

The concept of distance learning is not entirely new; its roots can be traced back several centuries. Historically, the desire to make education more accessible has driven innovation in the way knowledge is delivered. One of the earliest known forms of distance education dates back to the

18th century, when correspondence courses began to gain popularity. During this time, students would receive course materials through the mail, complete their assignments independently, and send them back to instructors for grading. While this method lacked real-time interaction, it provided an alternative for individuals who could not attend traditional institutions due to geographic, economic, or social constraints.

As the 20th century unfolded, technological advancements such as the radio, television, and telephone began to influence the way education was delivered. Educational broadcasts allowed institutions to reach wider audiences, and in some countries, televised lectures became a core part of national education strategies. For example, in the 1960s and 70s, several universities in the United Kingdom and the United States developed "open university" models, which combined printed course materials with radio and TV programming to provide structured learning experiences at a distance. This era marked a significant shift in how people perceived education it was no longer confined within the walls of a classroom.

1.1.2. CONCEPTUAL BACKGROUND

The development of a Virtual University platform is grounded in a broad conceptual framework that draws from both educational theory and the evolving relationship between technology and learning. At its core, this concept is rooted in the belief that education should be inclusive, flexible, and adaptive to the changing needs of learners in the 21st century. It is not just about digitizing lectures or assignments   it is about rethinking the very structure of how knowledge is accessed, delivered, and validated in a world that is increasingly shaped by digital experiences.

The idea behind online universities stems from the convergence of two important concepts: **e-learning** and **distance education**. E-learning refers to any form of learning facilitated by electronic technology, particularly over the internet. It includes virtual classrooms, interactive content, video lectures, discussion boards, and online assessments. Distance education, on the other hand, is a broader term that includes any educational process where the learner and instructor are not physically present in the same location. While distance education existed long before the internet, the integration of e-learning has made it more interactive, efficient, and scalable than ever before.

This platform is also built on the concept of **accessibility** in education. Traditionally, access to quality higher education has been constrained by factors such as geography, cost, and institutional capacity. Students from rural areas or low-income backgrounds often face insurmountable challenges when trying to attend reputable universities. The conceptual model for a Virtual University challenge this by removing the physical and financial barriers that have historically limited access. By providing all academic services from application to graduation online, the platform ensures that anyone with a basic internet connection can participate in higher education, regardless of their physical location or personal circumstances.

1.1.3. THEORETICAL BACKGROUND

The development of a Virtual University platform is not merely a technological innovation; it is also deeply rooted in various educational theories that have evolved over time to explain how people learn, what makes learning effective, and how learning environments can be structured to support different types of learners. This theoretical foundation provides the guiding principles behind the design, functionality, and pedagogical approach of the platform.

One of the most prominent theories relevant to this project is **Constructivism**, which argues that learners build their own understanding and knowledge of the world through experiences and reflection. According to theorists like Jean Piaget and Lev Vygotsky, learning is not a passive process where information is simply absorbed; instead, it is an active process of constructing meaning. The Virtual University platform embraces this idea by offering interactive learning tools, discussion forums, and assignments that encourage critical thinking and problem-solving. Rather than just reading or listening, students are encouraged to explore, collaborate, question, and reflect all of which are essential to a constructivist learning environment.

Closely related is **Vygotsky's Social Development Theory**, particularly the concept of the **Zone of Proximal Development (ZPD)**. This theory highlights the importance of social interaction and guidance in learning. According to Vygotsky, students can achieve higher levels of understanding when supported by peers or instructors a concept known as "scaffolding." In the Virtual University context, this is reflected in features such as peer-to-peer discussions, collaborative projects, instructor-led sessions, and live group tutorials. These elements ensure that learners are not isolated, but rather guided and supported throughout their academic journey.

Another key theory informing this platform is **Cognitive Load Theory**, proposed by John Sweller. This theory suggests that instructional design must consider the limits of working memory. Overloading learners with too much information at once can hinder understanding and retention. Therefore, the platform is structured to deliver content in a manageable and organized manner using chunked materials, clear navigation, progressive learning paths, and a mix of text, visuals, and video to cater to different learning preferences. By managing cognitive load effectively, the platform enhances comprehension and long-term retention.

1.1.4. CONTEXTUAL BACKGROUND

The development of a Virtual University platform does not occur in a vacuum. It is shaped and influenced by the broader context in which it is conceived and implemented including technological, educational, economic, and societal factors. Understanding this context is essential to fully grasp the relevance and timeliness of the platform, especially within the specific setting or country for which it is being built.

In recent years, there has been a growing recognition globally that traditional education systems are struggling to meet the demands of modern learners. Many universities, especially in developing countries, face issues such as overcrowded lecture halls, limited access to quality learning materials, underpaid teaching staff, and inadequate infrastructure. These challenges have created a gap between the supply and demand of quality education, leaving many students unable to access the learning opportunities they desire or deserve.

This problem is particularly evident in regions like Sub-Saharan Africa, South Asia, and parts of Latin America, where the youth population is rapidly increasing. In countries like Cameroon, Nigeria, Kenya, and others, the number of secondary school graduates far exceeds the capacity of available higher education institutions. As a result, many qualified students are either forced to defer their education, travel long distances to urban areas in search of schools, or give up altogether. Even for those who gain admission, the quality of instruction may be compromised due to outdated curricula, limited facilities, and overburdened teachers.

1.2. PROBLEM STATEMENT

Access to quality higher education remains a significant challenge in many parts of the world, particularly in developing regions where institutional resources are limited, infrastructure is inadequate, and student demand far exceeds capacity. Despite global recognition of education as a fundamental human right and a key driver of socio-economic development, millions of young people continue to face barriers that prevent them from pursuing university-level studies. These barriers range from financial constraints, physical constraints, and geographical limitations to rigid educational structures that are not compatible with the realities of modern life.

1.3. OBJECTIVES OF THE STUDY

1.3.1. MAIN OBJECTIVES

By the end of this work, the researcher will design a Virtual University Platform which will enable individuals seeking for education, but face challenges such as financial, physical, and geographical limitations to still be able to get quality education without them having to worry about all these constraints.

1.3.1. SPECIFIC OBJECTIVES

The studies include implementing this Virtual University Platform (or E-University) with the use of web application software using technologies such as Next.js, Node.js, CSS (tailwindcss), MongoDB, and many others in order to be able to ensure the effective rendering of the services required by the community and the students.

Subsequently, there will be an implementation of a mobile application for the students to be able to use for quicker-class-access.

Specific objectives of the web application include:

- Ensuring that students and teachers are able to gain or secure admission into the University by filling out the admissions form, submitting, and then it being reviewed and a response sent to them.

- Students and Lecturers should be able to receive email notifications in various scenarios for updates. For example, they should be able to receive email alerts when their application gets accepted or rejected.
- Lecturers should be able to upload assignments for their students to answer, and the students should be able to upload their assignment done as well all in PDF or Word document format (.docx or .pdf).
- Lecturers will be able to go on live class sessions with their students and deliver their courses, and the students will be able to attend these live sessions.
- The faculty or admin should be able to post announcements, and all teachers and students should be able to get these assignments on their platforms.
- Lecturers should be able to grade students' assignments, exams, etc., and the students be able to see their grades.
- Lecturers should be able to record their ongoing classes, which will later be stored in an archive so that students can later revisit those lessons or classes. The recorded classes will be automatically deleted one week after they have been recorded for storage management, giving students up to one week to either download or rewatch the video.

1.4. RESEARCH QUESTIONS

1.4.1. GENERAL RESEARCH QUESTIONS

Can putting in place a Virtual University Platform be an effective method for solving all the problems mentioned above?

1.4.2. SPECIFIC RESEARCH QUESTIONS

- How can a fully online university platform be designed to accommodate students from diverse geographic and socioeconomic backgrounds without compromising accessibility or quality?
- What core features and functionalities are essential for creating a seamless end-to-end academic experience in a virtual university setting (from application to graduation)?
- What challenges do students in remote or underserved regions face in accessing higher education, and how can an online university platform address these challenges?

- To what extent can online learning tools and strategies replicate or improve upon the academic outcomes of traditional, in-person university education?
- How can learning flexibility (such as asynchronous access and self-paced study) be effectively implemented without diminishing student engagement and accountability?
- What security and data privacy considerations must be accounted for when handling student records, exams, and other sensitive processes in a fully online academic platform?
- How can real-time communication and collaboration between students, instructors, and administrators be fostered in a virtual learning environment?
- What role does digital infrastructure (internet access, device ownership, digital literacy) play in determining the success of online higher education initiatives in developing regions?

1.5. RESEARCH HYPOTHESIS

Hypotheses are tentative solutions given to the research questions. These hypothesis guide research questions and provide a framework for empirical testing. These hypotheses typically articulate the expected outcomes or relationships between different elements within the Virtual University Platform or between the system and student and teacher's performance.

1.5.1. SPECIFIC HYPOTHESES

- There is a significant positive relationship between the accessibility of the online university platform and students' enrollment rates in higher education.
- The integration of real-time communication tools (e.g., chats, forums, live classes) on the platform significantly enhances student–instructor engagement.
- Students using the online university platform perform academically at the same level or better than those in traditional in-person university settings.
- There is a significant relationship between the flexibility of the platform (e.g., self-paced learning) and student satisfaction.
- The availability of a centralized, fully online application and admission process significantly reduces student dropout rates during the enrollment phase.
- Data security and privacy features on the platform influence the trust and willingness of students to submit sensitive academic information online.

- There is a significant correlation between students' digital literacy levels and their ability to navigate and succeed within the online university platform.

1.6. SIGNIFICANCE OF THE STUDY

The development of a fully online university platform stands to make a transformative impact on the landscape of higher education, particularly in developing nations where access, affordability, and infrastructure remain major barriers. This study is highly significant, not just as a technical contribution, but as a socio-educational innovation that addresses real-world challenges faced by students, educators, and institutions alike.

Students

For students, especially those living in remote or underserved regions, this project offers a practical and scalable pathway to access quality higher education without the need to relocate, commute, or bear the extra cost of living in urban centers. Many capable and driven individuals abandon their academic aspirations simply because they cannot afford tuition, transportation, or accommodation near traditional universities. This platform seeks to eliminate such hurdles by bringing the classroom directly to the student's device.

Institutions

This project also provides an opportunity for lecturers, tutors, and academic institutions to expand their reach and impact. Through the platform, educators can deliver high-quality content to students regardless of location, enabling broader dissemination of knowledge. The system promotes innovative teaching approaches such as blended learning, asynchronous instruction, and the use of multimedia teaching aids, all of which enhance the teaching experience.

Government

From a policy perspective, the findings and outcomes of this study can help inform government decisions on digital education strategies, educational equity, and the promotion of ICT in the academic sector. The system aligns with global efforts to digitize education and could support national goals related to increasing literacy rates, improving human capital development, and bridging the urban-rural education divide. As a low-cost yet high-impact initiative, it offers a scalable solution to the challenge of mass education without placing undue burden on physical infrastructure and government spending.

Community

For developers and researchers in the field of educational technology, this study presents a practical implementation model of a comprehensive e-learning ecosystem. It explores not only the technical aspects but also the human factors influencing the usability and effectiveness of digital learning systems. The knowledge gained from this project could inspire further studies on learning analytics, student engagement in virtual settings, system scalability, and data-driven personalization of learning content.

Society

On a broader scale, this study is significant because it champions the democratization of education making learning not a privilege of the few, but a right available to all. By removing geographic and economic barriers to higher education, the platform contributes to the creation of a more educated workforce, increased social mobility, and the long-term socioeconomic development of communities.

1.7. JUSTIFICATION OF THE STUDY

The justification for undertaking this study stems from the growing need to revolutionize the delivery of higher education, especially in regions where traditional education models are proving inadequate or inaccessible. This research is not just timely but necessary, as it directly responds to global trends, local educational gaps, and the evolving expectations of both learners and institutions in the digital age.

The Shift Towards Digital Education

Globally, there is an evident transition from brick-and-mortar learning environments to virtual platforms. The COVID-19 pandemic made this shift more urgent and widespread, but the benefits of online learning had already been recognized long before. What began as a temporary solution has now proven to be a viable long-term model. However, many developing countries, including those in Africa, still struggle to adopt fully digital learning systems due to technological, financial, and infrastructural constraints. This study is justified because it aims to provide a tailored, cost-effective, and sustainable online university solution that can bridge this divide.

Gaps in Accessibility and Inclusivity

A large number of students who graduate from secondary school never proceed to the university level due to limited admission slots, high tuition costs, long travel distances, or the absence of quality institutions in their locality. In rural areas especially, many capable youths are denied educational advancement purely based on geography. This study is crucial because it focuses on breaking down these barriers by proposing a system that allows students to apply, enroll, attend lectures, and even graduate all online.

Flexibility for Non-Traditional Learners

Education is no longer a domain exclusive to full-time students aged 18–25. Today, adults with jobs, parents with responsibilities, and individuals looking to switch careers also seek higher education. A rigid, campus-based learning structure often excludes such people. This research is justified as it accommodates this new breed of learners by providing a platform that supports self-paced learning, asynchronous lectures, and mobile access, all of which are essential for modern education.

Institutional Challenges and Inefficiencies

Many universities continue to rely on outdated, paper-based systems for admissions, course registration, assignment submission, and grading. These methods are prone to errors, inefficiencies, and delays. Moreover, they make data tracking, performance analysis, and decision-making quite difficult. This study addresses these challenges by proposing a centralized and digitized platform that simplifies administration, improves communication, enhances transparency, and reduces operational costs for institutions.

1.8. SCOPE OF THE STUDY

The scope of the study here involved **time, geographic, and thematic scope**

1.8.1. THEMATIC SCOPE

The project is titled a Full-Stack Virtual University Platform which is totally built with administrative privileges and only the administrator is given access to create, edit, and delete things.

The things that students and teachers are able to create are very limited and are well within the bounds of what students and teachers are normally supposed to be able to do. Nothing extra.

1.8.2. GEOGRAPHIC SCOPE

This project was carried out in the political capital of Cameroon, Yaoundé, precisely at Acasia, behind "Hotel Le Mendjang" at Eschosys Technologies

1.8.3. TIME SCOPE

This Project took over 3 months, from 29th June to 29th August; however, this period wasn't sufficient to carry out scientific research.

1.9. ORGANIZATION OF THE STUDY

- Chapter one which comprises of the general introduction, the background of study, problem of the statement, objective of the study, research questions, hypothesis, significance of study, justification and scope of study.
- Chapter two is concerned with the literature review which contains the theoretical review, conceptual review and empirical review
- Chapter three looks at the research methodology which includes the research design of the Full-Stack University Platform, data instrument analysis and interpretation.
- Chapter four is based on data presentation analysis and presentation.
- Chapter five is on the recommendation for future implementation and conclusion.

CHAPTER TWO

LITERATURE REVIEW

In this chapter, we are going to be seeing the following. Firstly, review by concepts, secondly, review by objective, and finally, presentation of internship activities.

2.1. THEORETICAL REVIEW

The theoretical review provides a structured foundation upon which this research is built. It examines the underlying theories, models, and frameworks that explain how and why an online university platform can function effectively, how users interact with technology in educational contexts, and what factors influence the outcomes of virtual learning environments. These theories help to validate the design, implementation, and expected results of the proposed system.

Constructivist Learning Theory

Constructivist theory, primarily associated with Jean Piaget and later expanded by Lev Vygotsky, asserts that learners actively construct their own understanding and knowledge of the world through experience and reflection. In a virtual university platform, this theory supports the idea of interactive, student-centered learning where knowledge is built through engagement with digital content, peer discussions, online activities, and self-paced exploration.

The online university fosters an environment where learners are not passive recipients but active participants in the learning process. Features such as forums, video lectures, virtual simulations, and collaborative projects reflect the core of constructivist thinking   learning by doing.

The Technology Acceptance Model (TAM)

Developed by Davis (1989), the Technology Acceptance Model explains how users come to accept and use technology. It is based on two key beliefs: **perceived usefulness** (the degree to which a person believes that using a system would enhance their performance) and **perceived ease of use** (the degree to which a person believes that using a system would be free of effort).

This theory is especially relevant to the adoption of the online university platform by students, teachers, and administrators. The system’s user interface, performance, and integration into daily academic activities must be designed in a way that ensures it is both useful and easy to navigate increasing the likelihood of acceptance and sustained use.

Self-Determination Theory (SDT)

Proposed by Deci and Ryan, SDT focuses on motivation and psychological needs. It emphasizes three core elements: **autonomy**, **competence**, and **relatedness**. In the context of virtual learning, SDT explains how students are more motivated to learn when they feel in control of their education (autonomy), believe they can succeed (competence), and experience connection with others (relatedness).

This theory justifies the inclusion of self-paced modules, progress tracking, feedback systems, and interactive communities within the platform all aimed at enhancing intrinsic motivation and academic success.

2.2. CONCEPTUAL REVIEW

The conceptual review delves into the key ideas, constructs, and principles that form the foundation of the virtual university platform. While the theoretical review focused on existing models and frameworks, the conceptual review defines the central concepts in the context of this research. It draws connections between core components such as e-learning, online student engagement, virtual learning environments, and the technology that powers them. This section helps establish a clear understanding of how the system operates and what elements are critical to its success.

Virtual Learning

Virtual learning refers to the use of digital technologies and internet connectivity to deliver education remotely. It breaks away from traditional face-to-face classroom experiences and instead leverages platforms that provide video lectures, assignments, quizzes, collaborative tools, and access to digital libraries. In a virtual university, learning is often asynchronous, meaning students access content at their own pace, although synchronous sessions (live classes or discussions) may also be available.

Learning Management System (LMS)

A Learning Management System is a software application that enables the creation, delivery, management, and tracking of educational content and student progress. It acts as the backbone of the virtual university platform. Through the LMS, instructors can upload course materials, create quizzes, hold discussions, and provide feedback, while students can access lessons, submit assignments, communicate with instructors, and monitor their academic performance.

Well-known LMS platforms include Moodle, Canvas, and Google Classroom. The virtual university platform proposed in this study will either integrate or be modeled after such systems, ensuring robust user experience, scalability, and educational value.

Online Student Engagement

Student engagement in an online setting includes how actively learners participate in their educational experience. It goes beyond simply attending virtual classes it includes contributing to discussions, completing assignments on time, collaborating with peers, and seeking help when needed. The concept of engagement is crucial in determining student satisfaction, retention, and academic success.

Digital Divide

The digital divide refers to the gap between individuals who have access to modern information and communication technology and those who do not. In the context of a virtual university, this is a critical concept because students in underserved or rural areas may face challenges such as poor internet connectivity, lack of digital devices, or low digital literacy.

Understanding the digital divide helps inform design decisions that make the platform lightweight, mobile-responsive, and adaptable to offline usage or low-bandwidth environments. It also reinforces the importance of inclusivity in the system's development and deployment.

E-Assessment

E-assessment is the process of using digital tools to assess student knowledge, skills, and performance. It includes online quizzes, timed tests, project submissions, and even AI-based proctoring systems. A well-structured e-assessment strategy provides immediate feedback, reduces administrative workload, and supports scalable evaluation for large numbers of students.

The virtual university platform must ensure that its assessment modules are secure, flexible, and aligned with academic standards while also providing options for both formative (continuous) and summative (final) assessments.

Student Information Systems (SIS)

A Student Information System is a digital database that stores and manages student records such as enrollment details, grades, attendance, and academic history. In a virtual university, the SIS works hand-in-hand with the LMS to provide a complete picture of a student's journey.

The platform in question will incorporate an SIS component that allows administrators and academic staff to manage student data efficiently, generate transcripts, track performance, and handle course registration digitally.

Communication and Collaboration Tools

Online education is only effective when communication channels are strong. This includes tools like messaging systems, email notifications, discussion boards, video conferencing (e.g., Zoom, Google Meet), and collaborative platforms like Google Docs or shared whiteboards. These tools help foster interaction between students and instructors, which is vital in reducing isolation and encouraging knowledge sharing.

Integrating these tools into the virtual university ensures that learning is not just a solitary activity, but one enriched by communication, teamwork, and real-time support.

Course Management and Content Delivery

Course management refers to the ability of instructors to create, organize, and schedule course content. Content delivery, on the other hand, involves the methods used to present this material to students such as recorded lectures, live sessions, downloadable PDFs, and interactive media.

The effectiveness of content delivery greatly affects learning outcomes. Therefore, the platform must prioritize ease of access, diversity of learning materials, and compatibility with various devices.

User Experience (UX) and Accessibility

User Experience refers to how users interact with and feel about a system. In educational platforms, poor UX can lead to student frustration, disengagement, or even dropouts. Accessibility, meanwhile, ensures that the platform can be used by individuals with disabilities for example, through screen reader support, keyboard navigation, and captioned videos.

Both UX and accessibility are central to the platform's design principles to make learning intuitive, inclusive, and enjoyable.

Data Privacy and Security

A virtual university must handle sensitive data from personal student information to academic records and financial details. The concept of data privacy and cybersecurity is essential to build trust and ensure compliance with legal standards such as GDPR or local data protection laws.

2.3. EMPERICAL REVIEW

The empirical review focuses on analyzing previously conducted studies, real-world findings, and tested implementations relevant to online learning platforms and virtual university systems. By reviewing empirical literature, we aim to understand what has already been observed, measured, or tested in the domain of online education, particularly as it relates to platform functionality, learner outcomes, user experience, and overall system success or failure.

Online Learning and Academic Performance

A substantial body of empirical evidence supports the effectiveness of online education when implemented with thoughtful instructional design and robust digital infrastructure. For instance, a 2018 study by Bernard et al. found that students enrolled in online courses performed similarly or better than those in face-to-face classes, especially when given access to interactive tools and instructor feedback. Their meta-analysis of over 100 studies revealed that online learners who engaged actively with course material and instructors were more likely to succeed.

Another empirical study by Means et al. (2010), commissioned by the U.S. Department of Education, analyzed over a decade's worth of research and concluded that students in online learning environments performed modestly better, on average, than those receiving face-to-face instruction. This suggests that under the right conditions, virtual learning can rival or exceed traditional models.

Access to Higher Education through Online Platforms

Empirical findings from UNESCO (2021) indicate that online learning platforms play a crucial role in extending access to education, particularly in underserved regions. During the COVID-19 pandemic, many countries observed increased enrollment in online education programs, and for some students, it was the only viable option to continue or begin their tertiary studies. This shift highlighted the potential of virtual platforms to bridge the educational gap between urban and rural populations.

User Experience and Platform Design

The usability and intuitiveness of online university platforms have also been widely studied. A research paper by Sun et al. (2008) emphasized that student satisfaction in online learning is highly dependent on factors such as system reliability, user interface design, clarity of content, and the instructor's responsiveness. Their findings stressed the need for platforms to be simple, mobile-friendly, and accessible to students with limited digital literacy.

Assessment and Evaluation in Virtual Learning

Effective student assessment is another area of concern in fully online universities. A 2021 study by Khalil and Ebner explored the credibility of online examinations using proctoring software. Their results showed that although students found the surveillance uncomfortable, it helped reduce academic dishonesty and increased trust in the certification process.

Similarly, empirical research in India and Pakistan indicated that systems with built-in real-time monitoring, question randomization, and AI-supported evaluation mechanisms made remote assessments more credible and fairer, especially in high-stakes examinations.

2.4. PRESENTATION OF THE ENTERPRISE

2.4.1. PRESENTATION OF INTERNSHIP ACTIVITIES

ESCHOSYS TECHNOLOGIES is a dynamic and innovative technology company dedicated to delivering a wide range of technological services. Through a team of skilled professionals, we specialize in various technological fields, offering solutions that cater to diverse client needs. This mission of ESCHOSYS TECHNOLOGIES is to enhance efficiency and productivity through cutting-edge technology, while our vision is to become a global leader recognized for quality and customer satisfaction. Coupled with the fact that ESCHOSYS is geared towards enhancing hands-on experience with their trainees, some of the core activities or services provided by ESCHOSYS TECHNOLOGIES are:

Application Development

ESCHOSYS TECHNOLOGIES specializes in building Mobile, Web, and Desktop application for its clients using a variety of different technology stacks

Graphic Design

Trainees and trainers are ESCHOSYS use adobe products like Adobe Photoshop, Illustrator, and InDesign, to create visually compelling designs for marketing materials, branding, and more.

2.4.2. ACTIVITIES CARRIED OUT

ESCHOSYS TECHNOLOGIES is a sun rising Tech Company that is made up of few but active departments. Some of the functional departments in ESCHOSYS are:

I.T (Information Technology)

This department serves as the live wire of ESCHOSYS. This is so because ESCHOSYS is a tech company that most deal with 17 tech solutions, services or products and all the tech personnel are highly involved in all the tech activities

Management

This department oversees all the different activities of the company and ensures that management is at its peak performance to guarantee the wellbeing and sustainability of the company.

Human Resource Management

This department is in charge of recruiting new trainers/professionals are recruited into the company based on severe recruitment procedures

2.4.3. INTERNSHIP EXPERIENCE

This part of the study is about the internship experience. It includes both professional and personal experience.

2.4.3.1. PROFESSIONAL EXPERIENCE

During my three months' internship at ESCHOSYS TECHNOLOGIES, the researcher learned so many things which the researcher wasn't aware of. The researcher mastered new application skills in the field of IT. These skills which the researcher will be able to use as a professional in order to solve problems or challenges in my community and society.

2.4.3.2. PERSONAL EXPERIENCE

The researcher personally enjoyed their time as an intern for ESCHOSYS TECHNOLOGIES. Due to the knowledge which, the researcher gained from there, the researcher was able to be more and more confident of myself and my capabilities as an aspiring Software Engineer. The three months internship period helped to occupy the researcher during holidays, to ensure that the researcher wasn't idle and keep his mind focused constantly.

2.4.3.3. GENERAL IMPRESSION AND WORKING CONDITION

This part of the work will be about the general impression and the working conditions at the internship place.

2.4.3.4. GENERAL IMPRESSION

During the internship period at ESCHOSYS TECHNOLOGIES, the interns were received by the manager of the company. Dr. Shiynsa Charles. The interns were then briefed on the various activities which are carried out in the organization. The employees where generally very welcoming, and made sure to answer any questions which were asked by interns and a little career orientation was carried out. So, overall, ESCHOSYS TECHNOLOGIES left a very good impression on the researcher.

2.4.3.5. WORKING CONDITION

The official working days at ESCHOSYS TECHNOLOGIES are from Mondays to Saturdays during internship periods and from Mondays to Fridays for the normal working period. Work starts at 8am followed by a 1hr break from 12pm to 1pm and classes training continues till 3pm.

2.4.4. STRENGTH AND WEAKNESSES

This part of the work contains the contains the strengths and weaknesses of ESCHOSYS TECHNOLOGIES.

2.4.4.1. STRENGHTS

Though ESCHOSYS TECHNOLOGIES is a sun rising Tech Company, there exist many merits that makes ESCHOSYS outweigh other tech companies. Some of these advantages that make the company powerful are:

Super-fast internet connectivity

ESCHOSYS provides super-fast starlink internet connectivity to its employees and interns, internet with speed of up to 300mbps, permitting all activities to be carried out smoothly without any delay, and also encourages learning.

Quality and Reliability

Ensuring that products and services meet the highest quality standards, ESCHOSYS also ensure that training offered to trainees and services rendered are of quality based on feedback and testimonies from customers. Reliability and Performance: they offer reliable and high-performance solutions that customers can depend on.

2.4.4.2. WEAKNESSES

Though there might be so many positive aspects about ESCHOSYS, there are also some few drawbacks that need adjustments. Some of these drawbacks are:

Marketing

At times the marketing team is very lax and this goes a long way to retard the visibility of the company and the services offered by the company as a whole.

Punctuality of trainers/Tech professionals

Provided that most of the tech professionals of ESCHOSYS develop software into the late hours of the night, they mostly come late to the company. This goes a long way to affect the stipulated objectives and the company objectives as whole and expected outcomes.

2.4.6. PROBLEMS ENCOUNTERED

Though ESCHOSYS TECHNOLOGIES was best of an internship place, there are also challenges encountered carrying out internship. Some of the challenges encountered are:

Internship fee

With the aim to purchase the necessary material needed by the trainers to transmit skills, there was need for a minimal amount as internship fee. This was still a problem to the intern as at given moments raising transport and raising the internship fee was a bit difficult.

Language challenges

Helping out fellow internship mates and friends with some difficulties because they could not master the language (English or French) some of the software were installed or the trainers were teaching is another challenge. This ending up with a good experience because modern AI tools that have aided translation and configuring of software. Communicating fluently with those of purely French background was also a challenge.

2.4.6. RECOMMENDATIONS

Some of the few recommendations as regards the various challenges and weaknesses encountered at ESCHOSYS TECHNOLOGIES are:

Assiduity

Trainers should always endeavor to be on time at the company. This will go a long way to meet company objectives and the internship program scheduled for the period available.

Marketing

The management of ESCHOSYS should cease from giving a blind eye to the laxity of the marketing department. If serious measures are not taken, the might loss popularity and visibility to the entire world provided is just a sun rising company.

CHAPTER THREE

METHODOLOGY

3.1. INTRODUCTION

This chapter provides a comprehensive explanation of the research methodology and materials employed in the development and evaluation of the Virtual University Platform. The methodology outlines the strategies, techniques, and procedures that were adopted to gather data, design and develop the system, and evaluate its functionality and effectiveness. It also explains the rationale behind the chosen methods, the tools and technologies used, and how the research objectives were systematically addressed.

The chapter begins by detailing the research design and approach, followed by a description of the population and sample involved in the study (if applicable). It then outlines the data collection techniques used both qualitative and quantitative as well as the instruments and materials employed for system development. The software architecture, programming languages, frameworks, and testing tools are also presented in this section.

3.2. DESCRIPTION OF THE ARCHITECTURE OF THE SYSTEM

The architecture of the Virtual University Platform has been carefully designed to meet the demands of a fully online academic environment, ensuring scalability, accessibility, security, and user-friendliness. The system follows a **three-tier client-server architecture**, which consists of the **presentation layer (front-end), the application layer (back-end), and the data layer (database)**. This modular structure enhances maintainability and allows the system to scale as the number of users grows.

Presentation Layer (Front-End)

This is the layer that interacts directly with the users, including prospective students, admitted students, instructors, and administrators. It is built using modern web technologies such as **HTML5, CSS3, JavaScript, and Next.js**, which provide a responsive, dynamic, and intuitive user interface. For mobile responsiveness and better user experience across devices, the front-end utilizes component-based design, making it easy to reuse and modify UI components.

The front-end is responsible for:

- Displaying application and enrollment forms
- Rendering course content and virtual classrooms
- Handling real-time interactions such as chats or notifications
- Ensuring easy navigation through various user roles (admin, student, lecturer)

Application Layer (Back-End)

The application logic, or the brain of the system, resides in the back-end. This layer was developed using **Node.js with Express**, offering a robust and scalable environment for managing data processing, authentication, API endpoints, and business logic. This layer is responsible for:

- Handling user registration, login, and role-based access control
- Managing courses, subjects, assignments, and examinations
- Coordinating communication between the front-end and the database
- Processing form submissions and admissions-related workflows
- Sending automated notifications and email alerts

The RESTful API structure adopted ensures clear separation of concerns and enables integration with third-party tools or services, such as payment gateways or external learning platforms.

Data Layer (Database)

At the core of the system is the **MongoDB** NoSQL database, which offers high flexibility and performance in handling large volumes of unstructured or semi-structured data. MongoDB is suitable for storing diverse user records, applications, course materials, attendance logs, and grades. Key database functionalities include:

- Storing and retrieving student and faculty information
- Managing course content and class schedules

- Tracking application statuses and admission decisions
- Recording academic performance and attendance

To enhance data security, sensitive fields such as passwords are encrypted using hashing algorithms, and role-based access controls restrict what each user can view or modify.

System Hosting and Deployment Architecture

The platform is hosted on a **Virtual Private Server (VPS)** running **Ubuntu Linux**, ensuring full control over deployment, configuration, and performance optimization. The server uses **Nginx** as a reverse proxy and **PM2** to manage Node.js processes. Continuous Deployment (CD) is handled via GitHub workflows, allowing for seamless code updates and version control. SSL encryption is enabled to ensure secure data transfer across all connections.

System Security and Authentication

Security is a critical part of the system architecture. The application uses **JWT (JSON Web Tokens)** for session management and user authentication. User roles (student, lecturer, admin) are strictly enforced throughout the application. Additionally, the system employs HTTPS, input validation, database query sanitation, and anti-CSRF mechanisms to guard against common web vulnerabilities.

3.3. DATA COLLECTION AND USERS' NEEDS

Here we are going to brief on a few methods of data collection (that is explain the ways in which our data was collected in order to design our system to meet user needs.

Although data can be valuable, too much information is unwieldy, and the wrong data is useless. The right data collection method can mean the difference between useful insights and timewasting misdirection. Luckily, organizations have several tools at their disposals for primary data collection. The methods range from traditional and simple, such as a face-to-face interview, to more sophisticated ways to collect and analyze data.

3.3.1. OBSERVATION

Observation involves collecting information without asking questions. This method is more subjective, as it requires the researcher, or observer, to add their judgment to the data. But in some circumstances, the risk of bias is minimal. For example, if a study involves the number of people in a university at a given time, unless observer counts incorrectly, the data should be reasonably reliable. Variable that requires the observer to make distinctions, such as how many millennials visit a university in a given period, can introduce potential problems. In general, observation can determine the dynamics of a situation, which generally cannot be measured through other data collection techniques. Observation also can be combines with additional information, such as videos. Below is what we observed during data collection for target system.

Paper-based grading systems

Lecturers manually record scores in books or spreadsheets, which leads to errors or data loss.

Delayed result publication

Students wait weeks or even months to access results due to slow result compilation and validation procedures.

In-person lectures only

Students must attend classes physically, with no option for recorded sessions or virtual learning creating a disadvantage for sick or distant learners.

Manual admission application and screening

Prospective students physically submit forms and wait for processing, which is slow and disorganized.

No centralized database for students

Information is scattered in multiple offices or physical files, making it difficult to track student progress or retrieve data on demand.

Lack of notification system

Students rely on bulletin boards or word-of-mouth for announcements, leading to missed deadlines and confusion.

3.3.2. INTERVIEWS

To gather accurate user requirements for the Virtual University Platform, structured and semi-structured interviews were conducted with three key user groups: **students**, **lecturers**, and **university administrators**. These interviews were crucial in identifying existing challenges, understanding user expectations, and guiding the design of a user-centric virtual learning environment.

The interviews aimed to:

- Understand the pain points in current academic and administrative processes.
- Identify gaps in learning delivery and academic support.
- Determine the willingness and capacity of users to adopt a digital platform.
- Gather feature suggestions and usability expectations for the virtual platform.

3.3.3. USER'S NEEDS

After conducting a series of in-depth interviews with various stakeholders involved in tertiary education including students, lecturers, administrative staff, and IT personnel as well as observing the traditional systems used in managing university processes, several critical needs were identified. These needs are at the heart of the development of this Virtual University Platform.

Firstly, it was observed that the traditional method of university application and admission is highly manual, time-consuming, and often plagued with inefficiencies. Students are required to physically travel to institutions to submit application forms, obtain receipts, and check results. This process is not only costly and stressful for applicants especially those coming from distant regions but also introduces a high potential for human error, document misplacement, and administrative delays.

Additionally, there is a major gap in how lectures are delivered and how learning resources are distributed. In many traditional setups, students depend almost entirely on physical presence in lecture halls and handouts provided by lecturers. If a student misses a class due to illness or other valid reasons, they often find it difficult to catch up. Moreover, there is limited access to centralized digital course materials or recorded lectures, making revision difficult and uneven across the student body.

3.4. FUNCTIONAL REQUIREMENTS

Functional requirements define the specific behaviors, features, and functionalities that the system must support to meet user expectations and fulfill its intended purpose. These are the core features that allow the Virtual University Platform to function efficiently as an end-to-end academic management system, serving students, lecturers, and administrators alike.

After careful analysis of user needs, observations, and interviews, the following functional requirements have been identified for the Virtual University Platform:

User Registration and Authentication

- The system must allow new users (students, lecturers, admins) to create accounts with appropriate roles and access levels.
- It must provide secure login mechanisms with email/password and optional two-factor authentication.
- Users must be able to reset or recover forgotten passwords securely.

Online Application and Admission Processing

- Prospective students must be able to apply to academic programs online by filling out an electronic form, uploading necessary documents, and tracking application status.
- Administrators should be able to review, approve, reject, or request more information from applicants.
- Admission letters should be auto-generated upon approval and downloadable by the student.

Learning Content Management

- Lecturers should be able to upload lecture notes, videos, assignments, and reading materials.
- Students should have access to these resources anytime via a centralized dashboard.
- Materials should be organized by semester, course code, and module.

Live and Recorded Classes

- The platform should integrate a video conferencing tool (e.g., Zoom, Jitsi, or custom WebRTC) to allow real-time lectures.

- Lectures should be recorded and stored so that students can revisit them anytime.

Continuous Assessment and Grading

- Lecturers must be able to create quizzes, tests, assignments, and projects within the system.
- Students should be able to take these assessments online with support for timed tests and automated grading where applicable.
- Lecturers should be able to manually enter grades and feedback, and students should view their scores in real-time or upon release.

Communication and Notifications

- The system must allow students and lecturers to communicate via an internal messaging system or email integration.
- Notifications (e.g., new grades, course announcements, system updates) should be sent via email and/or in-app.

Role-Based Access Control

- The platform must enforce strict access control depending on the user role (student, lecturer, admin, visitor).
- Each role must only access the features and data relevant to their permissions.

3.5. FUNCTIONAL SPECIFICATIONS

The **functional specifications** outline the precise technical and behavioral details that the system must implement in order to satisfy the functional requirements described earlier. These specifications go beyond the "what" of the system and dive into the "how" describing how each feature should behave, how users interact with it, and what constraints or conditions apply. Each module or component is broken down into its core functionalities, user actions, data processing flows, and system responses.

Below is a comprehensive breakdown of the major modules and their corresponding functional specifications:

User Authentication Module

- **Login Functionality**:
  - Input: Email and password.
  - Process: Credentials are checked against stored, hashed values in the database.
- **Registration Functionality**:
  - Input: Name, email, password, role (student, lecturer, admin), other profile details.
  - Process: Input validation, uniqueness check for email, password hashing.
  - Output: Account is created, and a confirmation email is sent.
- **Password Reset**:
  - Process: Email-based reset link with token validation.
  - Security: Tokens must expire after 15–30 minutes and be single-use.

Online Application System

- **Application Submission**:
  - Input: Applicant's personal data, academic background, documents (PDF, JPG), and program selection.
  - Process: Form validation, document upload, and entry stored in the admissions database.
  - Output: Unique application ID generated and visible to the applicant.
  - Admin View: Displays all submissions, sortable and filterable.
- **Admission Status Tracking**:
  - Input: Application ID.
  - Output: Displays status as *Pending*, *Under Review*, *Approved*, or *Rejected*.
  - Notification: Email alerts sent upon status change.

Learning Content Delivery

- **Content Upload (Lecturers)**:
  - File support: PDFs, DOCX, PPTs, MP4s, and embedded YouTube links.
  - Upload flow includes title, course code, and module/topic.
  - Content is searchable and grouped by week or module.
- **Content Access (Students)**:
  - Students view or download files.
  - System logs access for analytics and feedback.

Live Class Integration

- **Class Scheduling**:
  - Lecturers create live class events with date, time, course, and meeting link.
  - Students are notified via email and dashboard alert.
- **Recording and Playback**:
  - Classes are recorded (if enabled) and saved automatically.
  - Students can access the recording from the course page.

Assessment and Grading System

- **Assessment Creation**:
  - Lecturers create quizzes/tests with multiple-choice, short answer, or file upload formats.
  - Includes question banks and time limits.
- **Grading and Feedback**:
  - Auto-grading for objective questions.
  - Manual grading interface for subjective responses.

    - Students receive grades and inline feedback via the portal.

Result Computation and GPA System

- **Grading Scheme Configuration**:
    - Admin defines grade boundaries, weightings, and course credit values.
- **Result Display**:
    - Each student can view semester breakdown and GPA/CGPA summary.
    - Option to download as PDF transcript or shareable academic report.

Messaging and Notification Center

- **Inbox System**:
    - Students and lecturers can exchange direct messages or reply to course threads.
    - Admin can broadcast announcements to selected users.
- **Notification Alerts**:
    - Triggered on events such as: grade release, assignment deadlines, course registration periods.
    - Appear via email and in-app.

Payment and Billing System

- **Invoice Generation**:
    - System generates dynamic fee invoices based on student's program, level, and semester.
    - Students can view breakdown: tuition, administrative, and library fees.
- **Payment Processing**:
    - Supports payment via credit/debit card, mobile money, or bank transfer (API integration).

- Generates PDF receipt and updates dashboard balance.

Attendance Monitoring System

- **Automatic Tracking**:
  - For online classes, attendance is marked by system login duration.
  - For physical or manual sessions, lecturers tick present/absent status.
- **Student View**:
  - Summary of attendance percentage for each course, with color-coded indicators.

Feedback and Complaint Submission

- **Feedback Form**:
  - Allows rating of courses, lecturers, or general platform experience.
  - Includes open-text feedback, all submitted anonymously if desired.

Admin Dashboard and Control Panel

- **User Management**:
  - Create, update, suspend, or delete accounts.
  - Role assignment and permissions handled dynamically.
- **Academic Session Configuration**:
  - Admins define semesters, course listings, credit rules, and calendars.
- **System Analytics**:
  - Displays metrics such as user engagement, pass rates, most-accessed resources, etc.

Search and Filtering Functionality

- Search bars and filter dropdowns must be present on all list views (courses, grades, users, content).
- Results should support live search and return paginated data with optimized speed.

System Maintenance Tools

- Includes backup and restore options, database integrity checks, and user activity logs.
- Only accessible to top-level admins.

3.5.1. ROLE PLAYED BY EACH ACTOR IN THE SYSTEM

In the design and implementation of the Virtual University Platform, several actors are involved, each playing a vital role in ensuring that the system functions as intended. These actors interact with the system based on their specific responsibilities and privileges, and their activities collectively define the scope and performance of the platform. The major actors in the system include: **Students, Administrators, Lecturers, Admission Officers, and Technical Support Staff**.

Students

Students are the primary beneficiaries of the platform. Their role is crucial since the system is built to serve their educational needs. The following are the key activities performed by students within the platform:

- **Account Registration and Profile Management**: Students can sign up, fill in personal information, and update their academic profiles as needed.
- **Application Submission**: They submit online applications for admission into various programs. This includes uploading relevant documents and selecting preferred programs.
- **Receiving Admission Feedback**: After applying, students can monitor the status of their admission and receive notification of acceptance or rejection.
- **Attending Online Classes**: Students can participate in live video lectures or watch pre-recorded classes, download course materials, and interact with instructors.
- **Taking Examinations and Submitting Assignments**: The platform allows students to take timed online tests, submit assignments, and receive feedback.
- **Monitoring Academic Progress**: They can view their academic records, track GPA, and receive semester reports.

- **Communication and Support**: Students can message lecturers, join discussion forums, or submit support tickets when experiencing technical issues.

Administrators

The admin users are in charge of managing the platform's backend operations. They ensure that all modules and user accounts function correctly and securely. Their main tasks include:

- **User Account Management**: Creating and managing student, lecturer, and staff accounts.
- **Content and Data Management**: Monitoring content uploads, maintaining the course catalog, and archiving academic records.
- **System Monitoring and Maintenance**: Ensuring smooth performance of the platform, conducting system backups, and monitoring traffic.
- **Security Enforcement**: Setting access controls, updating roles and permissions, and handling system audits.
- **Resolving Escalated Issues**: Addressing technical complaints that could not be resolved by the support team.

Lecturers/Instructors

Lecturers are responsible for the academic delivery of content. They play an instructional role and interact directly with the students. Their roles include:

- **Delivering Lectures**: Hosting live classes through integrated video conferencing tools or uploading pre-recorded sessions.
- **Assigning and Grading Assessments**: Setting quizzes, assignments, and exams, then evaluating submissions and providing grades.
- **Student Interaction**: Responding to student questions, hosting virtual office hours, and moderating discussions.
- **Tracking Performance**: Monitoring student attendance, participation, and progress in their courses.

Admission Officers

Admission officers are tasked with handling the student intake process. They work closely with the applications submitted via the platform. Their responsibilities are:

- **Verifying Application Details**: Checking uploaded documents, verifying candidate eligibility, and filtering applications.
- **Admission Processing**: Making decisions on admissions and assigning admitted students to programs.
- **Communicating Decisions**: Updating the system with acceptance/rejection notifications and handling admission-related inquiries.

Technical Support Team

This team ensures that all users have a seamless experience when using the platform. Their roles include:

- **Providing Helpdesk Services**: Responding to queries, resolving login issues, and guiding users through various platform features.
- **System Bug Fixing**: Addressing reported technical bugs and deploying patches.
- **Training and Documentation**: Creating user guides, conducting training for students and staff, and updating FAQs.
- **Monitoring System Health**: Running diagnostics and performance monitoring to ensure system uptime and responsiveness.

3.5.2. FUNCTIONALITIES OF THE SYSTEM

User Authentication and Role-Based Access Control

- Users (students, lecturers, admins, and admission officers) can register, log in, and access the platform securely.
- The system uses role-based access control, meaning each user only has access to features appropriate for their role.
- Sessions are managed to prevent unauthorized access, and password recovery functionality is included.

Online Admission Management

- Prospective students can complete an online application form, upload required documents, and select preferred academic programs.
- Admission officers can review applications, verify uploaded documents, and make admission decisions directly within the system.
- Students receive notifications about the status of their applications in real-time.
- Automatic generation of admission letters upon approval.

Virtual Classroom and E-Learning

- The system integrates tools for delivering lectures online via live streaming (e.g., Zoom, Google Meet) and recorded content.
- Lecturers can upload lecture notes, video recordings, slides, and supplementary materials.
- Students can participate in online discussions, ask questions in forums, and access lecture archives at their convenience.

Assignment and Exam Management

- Lecturers can create and assign homework, quizzes, and exams with due dates and instructions.
- Students can submit assignments online, and the system can restrict late submissions or flag them.
- Automatic and manual grading features are available, and students receive timely feedback on their performance.
- Secure online exam module with time restrictions, question shuffling, and cheat prevention mechanisms.

Academic Performance and Transcript Generation

- The system automatically calculates and displays student grades, GPAs, and cumulative results based on predefined grading rules.

- Students can view academic history, semester results, and progress reports.
- Administrators can generate official academic transcripts in downloadable PDF format.

Communication and Notifications

- Built-in messaging system for communication between students, lecturers, and administrators.
- System-generated notifications for upcoming classes, assignments due, announcements, and admission decisions.
- Email and/or SMS integration to deliver critical alerts to users even when offline.

Administrative Control Panel

- Admins have access to a dashboard that gives an overview of user activity, course engagement, and system status.
- They can create, edit, and delete user roles, update academic programs, and assign instructors to courses.
- Logs of system activity for auditing and transparency.

3.6. TECHNICAL SPECIFICATIONS

The Virtual University Platform is a robust, scalable, and secure online academic system that integrates multiple technologies to deliver a smooth and reliable user experience. The technical specifications outlined in this section detail the hardware, software, development environment, programming languages, frameworks, databases, and hosting requirements necessary for the effective implementation and deployment of the system.

System Architecture

- **Architecture Style**: Client-Server (Web-based, RESTful API)
- **Tier**: Three-Tier Architecture (Presentation Layer, Business Logic Layer, Data Access Layer)
- **Deployment**: Cloud-Based Deployment using VPS

Front-End (Client-Side)

- **Technologies**: HTML5, CSS3 (TailwindCSS), JavaScript
- **Frameworks/Libraries**: React.js or Vue.js (for dynamic UI rendering)
- **UI/UX Tools**: Tailwind CSS for responsive and modern user interface
- **Browser Compatibility**: Chrome, Firefox, Safari, Edge (latest versions)

Back-End (Server-Side)

- **Programming Language**: Node.js (JavaScript)
- **API Type**: RESTful API for data access and operations
- **Authentication**: JWT (JSON Web Tokens) or Session-based Authentication with Role-based Access Control (RBAC)

Database Management

- **NoSQL**: MongoDB

Hosting and Deployment

- **Web Server**: Nginx
- **Operating System**: Ubuntu 20.04 LTS
- **Hosting Environment**: VPS
- **CI/CD Tools**: GitHub Actions, Jenkins for CI, automated testing and deployment
- **Domain & SSL**: Domain Name with HTTPS using Let's Encrypt SSL Certificate

Storage and File Management

- **File Uploads**: Local file system
- **Supported Formats**: PDF, DOCX, JPEG, PNG for student submissions, ID uploads, transcripts, etc.
- **Storage Structure**: Logical separation per user/role for secure file handling

Notifications & Messaging

- **Email Notifications**: SMTP integration (Nodemailer)
- **In-App Notifications**: Real-time toast or bell icon alerts using the websocket.io library

Security Specifications

- **Data Encryption**: SSL/TLS for data in transit, hashed passwords (bcrypt) for data at rest
- **Input Validation**: Sanitization at both client and server side to prevent XSS and SQL Injection
- **Access Control**: Role-based permissions and token/session expiration
- **Backup Strategy**: Daily automated database and file backups stored securely

Performance and Scalability

- **Caching**: Redis or Memcached for speeding up frequently accessed data
- **Load Handling**: Load balancing with Nginx
- **Scalability Plan**: Horizontal scaling for backend services and database read replicas when needed

Compatibility and Responsiveness

- **Device Compatibility**: Fully responsive design to support desktops, laptops, tablets, and smartphones
- **Cross-Platform Support**: Compatible with all major operating systems (Windows, macOS, Linux, Android, iOS)

Development and Testing Tools

- **Version Control**: Git with GitHub or GitLab
- **IDE/Editors**: VS Code
- **Testing Frameworks**: Jest (React/Node)
- **Linter and Formatter**: ESLint, and Prettier.

- **Docker Support**: Containerization for development and deployment consistency

3.7. NON-TECHNICAL SPECIFICATIONS

While technical specifications describe the underlying infrastructure and software components that support the functioning of the system, non-technical specifications address the broader expectations, constraints, and quality standards the system must meet to satisfy user needs, institutional goals, and operational environments. These specifications help ensure the system is usable, accessible, maintainable, and aligned with its intended purpose beyond just technical performance.

Usability

The system is designed with a strong focus on usability to accommodate a diverse group of users, including students, lecturers, and administrators with varying levels of technical expertise. The platform must:

- Provide an intuitive and clean user interface with clear navigation.
- Support both desktop and mobile users, ensuring accessibility on common devices.
- Allow users to accomplish their tasks (such as course enrollment, class attendance, or grading) with minimal effort or training.
- Include help guides, tooltips, and prompts where necessary to assist users.

Accessibility

Given the nature of the platform, it must be accessible to users across different regions, including those with limited digital literacy or users with disabilities. The system should:

- Comply with basic accessibility standards (e.g., WCAG 2.1).
- Provide screen reader support and keyboard navigation options.
- Allow content resizing and high-contrast modes for visually impaired users.
- Minimize reliance on high-speed internet connections by optimizing content delivery.

Maintainability

The platform should be designed in such a way that future updates, bug fixes, or feature additions can be implemented without disrupting the existing services. It should:

- Use modular code structures that make maintenance easier.
- Be well-documented, with clear descriptions of logic, modules, and data flow.
- Support easy backup and restoration of system data.
- Allow error logging and reporting mechanisms for troubleshooting.

Reliability

To maintain trust and academic integrity, the platform must be reliable. That includes:

- Minimizing system downtimes, especially during exams or live sessions.
- Ensuring data persistence (student records, coursework, etc.) and preventing accidental loss.
- Implementing robust error handling to prevent system crashes.

Security and Privacy

The system must safeguard sensitive information such as personal data, academic records, login credentials, and financial details. Therefore, it must:

- Ensure password encryption and secure authentication methods (e.g., two-factor authentication).
- Prevent unauthorized access through role-based permissions.
- Protect user data against breaches using secure coding practices and data encryption.
- Comply with privacy regulations, such as data minimization and informed consent for data usage.

Scalability

As the platform may be adopted by multiple institutions or experience an increasing number of students and faculty over time, it should:

- Be scalable horizontally and vertically, i.e., support more users without performance degradation.
- Allow for easy integration with third-party tools or platforms (e.g., video conferencing APIs, payment gateways).
- Support modular addition of features without having to redesign the core system.

Performance

Performance expectations include:

- Fast page loading times even on low-end devices or slower connections.
- Minimal latency in real-time features such as live classes or messaging.
- Efficient database queries to prevent lags during data-heavy operations like viewing grades or uploading assignments.

Legal and Ethical Compliance

The system must operate within the boundaries of legal and ethical expectations by:

- Adhering to local and international laws related to data protection and digital service delivery.
- Respecting copyright for uploaded course materials.
- Preventing academic dishonesty through features like proctored exams or plagiarism checks.

3.8. RESEARCH DESIGN

The research design for this virtual university platform is intentionally crafted to address the complications arising from the traditional, face-to-face method of managing university applications, student admissions, and course delivery. It aims to solve the following challenges:

- The difficulty and cost associated with traveling long distances to submit university applications or attend classes.

- Time wasted by administrative staff manually processing applications and managing academic records.
- Lack of a centralized and reliable platform for both students and instructors to access academic services and resources.
- The limitation of access to higher education for students living in remote areas or those with mobility constraints.
- Inefficiencies and delays caused by paper-based procedures and physical documentation.
- Inconsistent communication between students, lecturers, and administrative personnel.
- The absence of a fully integrated system that handles everything from application to graduation in a virtual environment.

This research design is therefore focused on creating a user-centered solution that ensures accessibility, efficiency, and reliability in delivering higher education completely online.

3.9. ANALYSIS METHODS

The analysis method are processes which make it possible to formalize the preliminary stages of the development of an information system in order to make this development more fruitful to the users' requirements. We will successively study the object-oriented methods and the functional methods.

3.10. OBJECT ORIENTED METHODS

The object-oriented method (OOMs) describes the static structure of the objects, their classes and their relations. One can mention here the following OOMs: OMT method, UML method and UP.

3.10.1. OMT METHOD

The object Modelling Technique (OMT) is an object modelling method for software modelling and designing. It was developed around 1991 by Rumbaugh, Blaha, Premerlani, Eddy and Lorensen as a method to develop object-oriented systems and to support object-oriented programming (ESPINASSE, 1980). OMT was developed as an approach to software development.

- The purposes of this modelling according to Rumbaugh are:

- Testing physical entities before building them(simulation);
- Communication with customers
- Conception (alternative presentation of information);
- Reduction of complexity

3.10.2. UML METHOD

UML is a language of modelling unified object in an object-oriented environment develop in response to the call for the proposal launched by the Object Management Group (OMG) with the goal of defining the standard notation for modelling of application built using. The principal authors are Grady Booch, Ivvar Jacobson and Jim Rumbaugh.

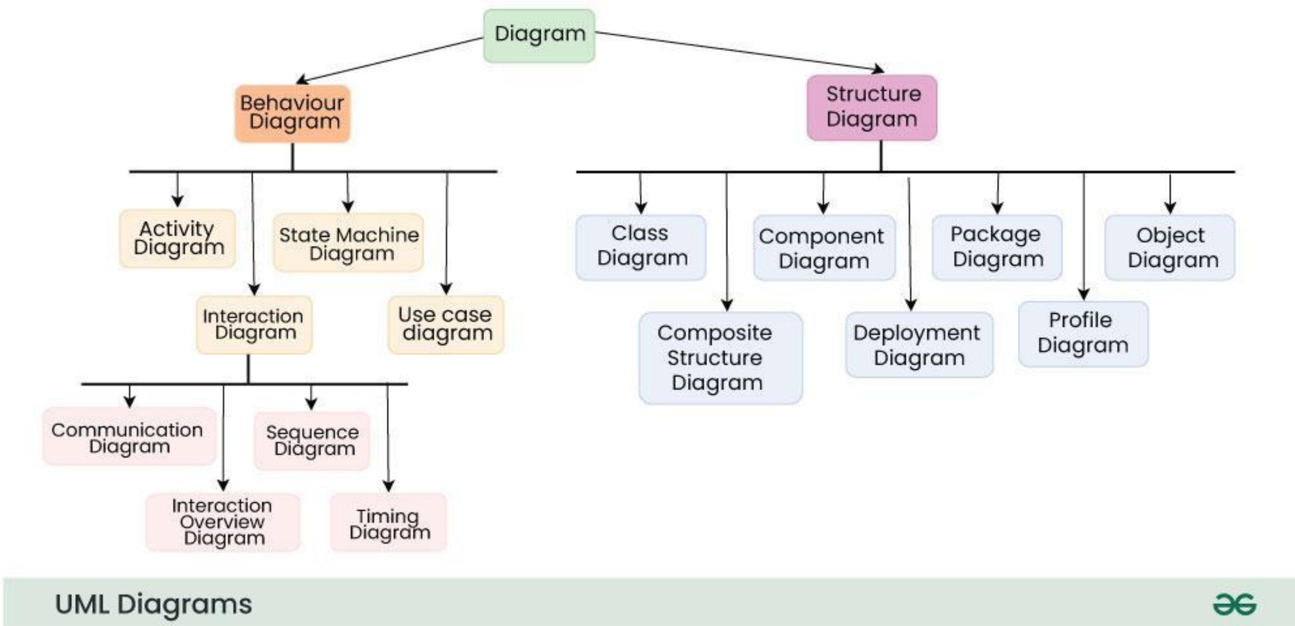

Figure 1: Overview of UML (source: GG)

Some advantages of UML are:

- Formal and standardized language, it allows proceeds of precision and constitutes a pledge of stability. This is what encourages the use of the tools.
- Powerful support of communication.
- Implementation of all the richness of the object approach.
- Description of all the models from the analysis to the realization of the software.

- Standardization of the concepts of objects.

Some limits of UML are:

- The semantics of UML is not formalized. It is specified by using the natural language.
- Difficulties optimization of the choice of classes.
- Various categories of diagrams are no formalized.

3.10.3. UP METHOD

Unified Process (UP) is a management method in the life cycle of software development and thus for object-oriented software. This is a generic method, iterative and incremental unlike the sequential method MERISE or SADT. This method is the general precept methods with the abbreviation: RUP, UPA XUP, EUP, 2TUP, AM, DCU. Thus, an embodiment according to UP, to transform the software needs of the users, must necessarily have the following characteristics:

- UP is based on components.
- UP uses UML.
- UP is driven by use cases.
- UP centric architecture
- UP is iterative and incremental.

Some advantages of UP are:

- Use case sensitive
- Architecture centric
- Iterative and incremental.

Some limits of UP are:

- It is used only at the beginning of the whole process to create business requirements.
- The final application reflects the business processes, but the exist no closer bond between them.
- A small change in the business process leads to a fundamental change of the created information system.

3.11. FUNCTIONAL METHODS

The functional methods have their origin in the development of the procedural languages. More towards the managements than towards data, they highlight the functional to be ensured and propose a hierarchical, downward and modular approach by specifying the bonds between the various modules. With the evolution of systems and programming languages, these methods took into account the modelling of data and the problems arising from real time.

3.11.1. SADT METHOD

Structure Analysis and Design Technique (SADT) Method is a method of American origin developed in 1977 by DOUG ROSS then introduced in Europe since 1982 by Michel GALINER. It is a multi-field language which supports the communication between users and originators. As a method of functional analysis and the most management of projects, SADT presents strong and weak points.

Some advantages of SADT are:

Its simplicity

- Its adequacy to capture the user's needs.
- Its capacity with being able to produce solutions on several levels of abstraction.

Some limits of SADT are:

- Its analysis is concentrated much on the functions, the coherence of data being neglected
- The rules of decomposition are not explicit. The decomposition differs according to analysts.
- Its difficulties of taking account of the non-hierarchical interactions in the complex system.
- Lastly, the volatility of the functions makes that the system is in perpetual Design.

3.11.2. MERISE METHOD

The MERISE (Method d'Etude et de Realisation Informatique pour les Systems d'Entreprise) method was launched around 1977 through a national consultation launched by the French Ministry of Industry with the aim to create a company of data processing consultant in order to define a method of design of information system. The MERISE method is based on separation of data and treatments to be carried out in several conceptual and physical models. The MERISE method recommend three levels of abstraction; **the conceptual level, the organizational level and the physical level.**

The conceptual level

The conceptual level defines the finalities of the company. It is on this level that objective to reach the constraints which weigh on the company are identified. It generally constitutes the most stable level and first level of development. At the conceptual level, one distinguishes the Conceptual Data Model (CMD) and the Conceptual Treatment Model (CTM).

The Organizational Level

The organizational level describes the organization which it is desirable to be set in the company to achieve the laid down objectives. The purpose of it is provide a diagrammatic representation of the organization of the company. One has heard of the Logical Data Model (LDM) and the organization and the Organizational Treatment Model (OMT). The organizational level is less stable and constitutes the second level of invariance. The physical describes the means which will be implemented to manage the data and to activate the treatments. It organized around the Physical Data Model (PDM) and the Operational Treatments Model (OTM).

3.12. CHOICE OF METHOD

Research on this work presents: OMT, UML, UP, SADT, MERISE as some of the principal models that can be used in designing an application. As a methodology to be used in this work, UML has been chosen to design our application. Automatically, UML will use the UP method because UP uses UML notations. The reason why UML is chosen is because in UML, the dynamics(behavioral) and static(structural) things are fused in to the system's entity to realize good and desirable results. This creates interdependency between the static and the dynamic things. It also provides precision and stability of the system. Hence, it is faster in building our application using UML to MERISE method. The MERISE method on the other hand, separates static approach system from the dynamic approach. It uses data models in representing the static system and treatment models in representing the dynamic system, it is not method made specifically for software development like UML but rather, it (MERISE) is generally used thus making the building of the application slower and costly because more materials are used to attain the same but less reliable result in quality and quantity.

3.13. APPLICATION OF METHOD (UML)

As it is often said, a picture is worth a thousand words, this absolutely fits while discussing about UML. UML is a pictorial standard and modeling mechanism for specifying, visualizing, constructing and documenting the artifacts of software systems. So beyond reasonable doubts, UML will help us better realize our application and understand its functionality with ease.

3.13.1. ACTORS

An actor is an external human or Software agent closer to the system and interacts with that system by its role played in our system. In case we have the following actors:

Admin

The admin manages the entire system, including user accounts, academic settings, and platform configurations.

Students

- Apply for admission
- Attend Online Classes
- Take Assessments
- View results

Lecturers

- Seek for admission
- Upload course materials
- Conduct lectures
- Assign and grade assessments
- Monitor students' progress.

3.13.2. DIAGRAMS

UML CLASS DIAGRAM

The class diagram is a static. It represents the static view of an application. A collection of class diagram represents the whole system. Class diagram is not only used for visualizing, describing and documenting aspects of the system but also for constructing executable code of the software application.

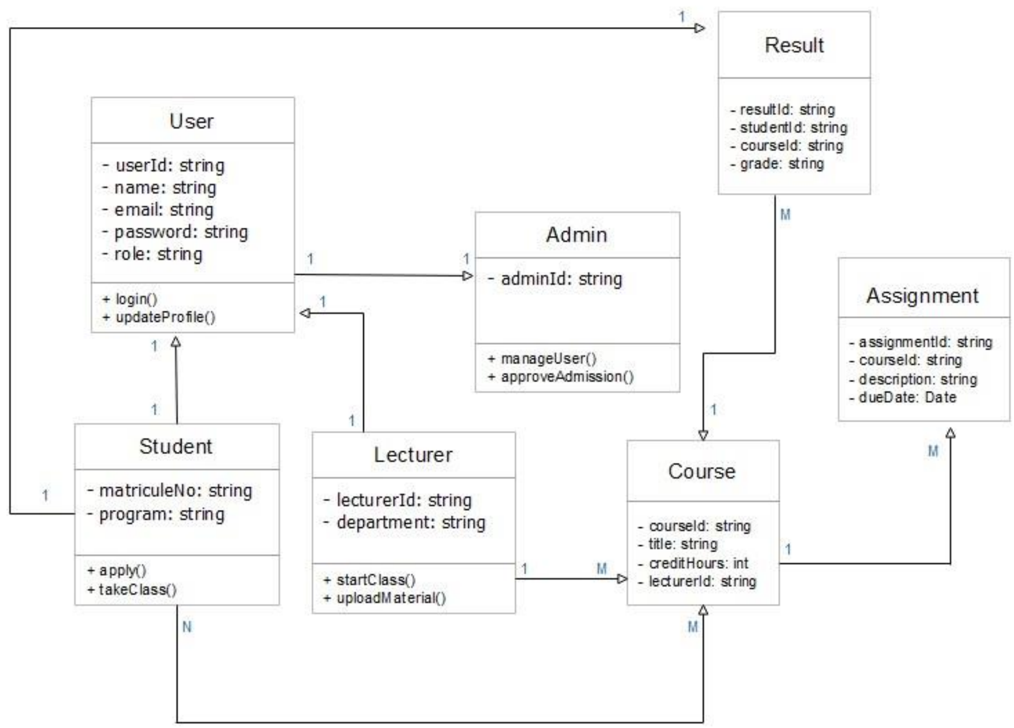

Figure 2: UML Class Diagram

UML USE CASE DIAGRAM

The Use Case or User case diagram is one which clearly shows all the actors in a given system and how those said actors interact with the system (their use cases).

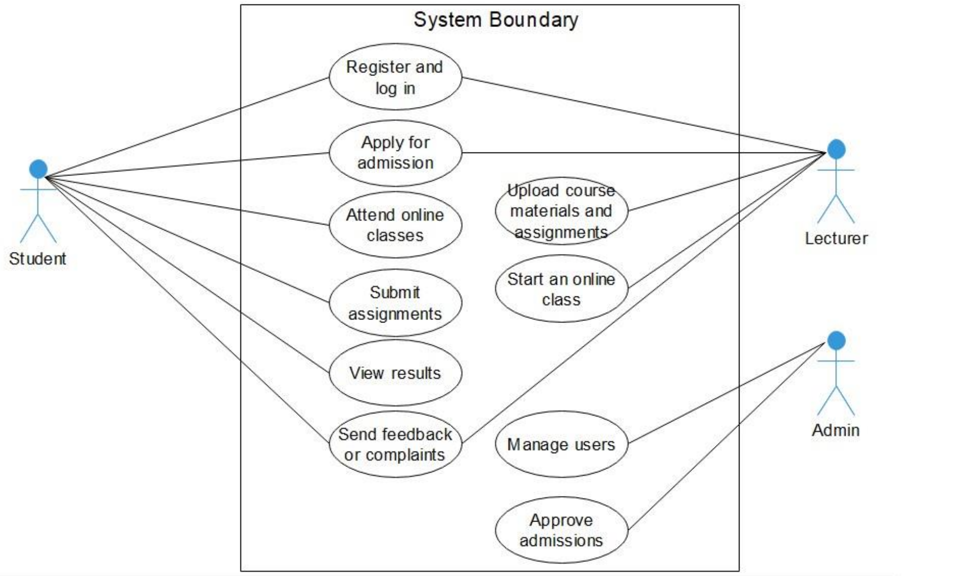

Figure 3: UML Use case diagram

UML SEQUENCE DIAGRAM

A sequence diagram is a type of UML (Unified Modeling Language) diagram that shows how objects or components in a system interact over time to complete a specific process or function. It shows how each actor interacts with another.

Figure 4: Sequence diagram

3.13.3. COMPONENTS OF THE SYSTEM

This Virtual University Platform is structured into several core components, each responsible for handling specific functionalities within the system. These components work together to ensure a seamless and fully online academic experience for all users.

Authentication and User Management

Handles user registration, login, logout, password recovery, and role-based access. It ensures that each user (student, lecturer, or admin) accesses only the features they are authorized to use.

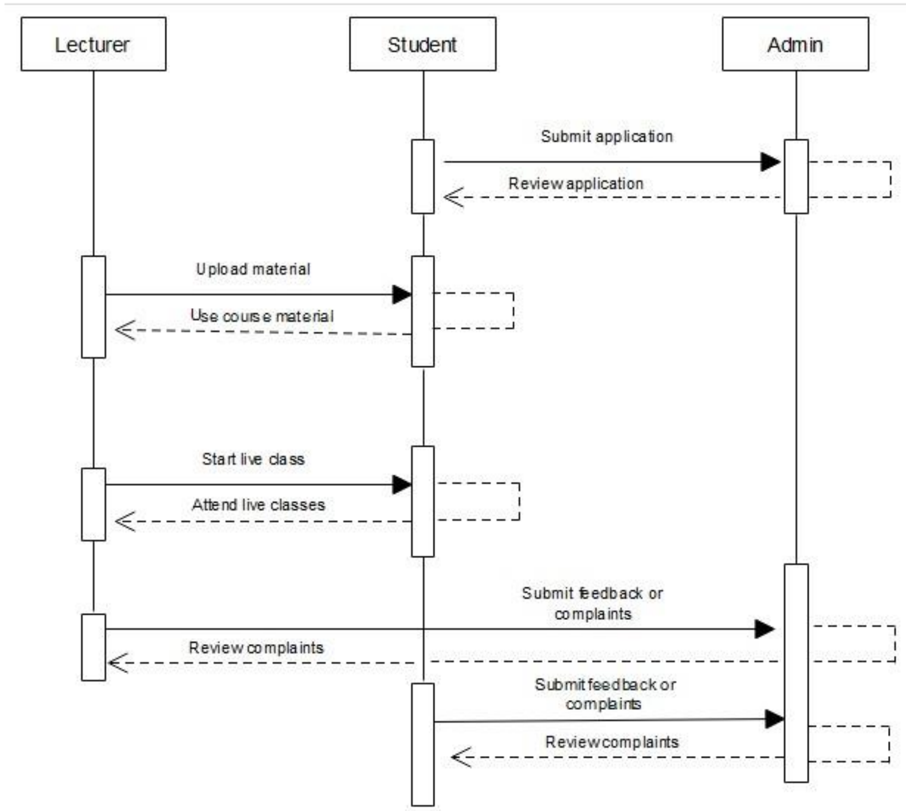

Figure 4: UML sequence diagram

Admission Management

Allows students to submit applications online and enables administrators or admission officers to review and process them. It includes document upload, program selection, and notification features.

Online Learning

Facilitates the delivery of live or recorded lectures, sharing of course notes, video playback, and discussion forums. It supports a fully virtual classroom experience.

Assignment and Exam

Allows lecturers to post assignments and exams, and students to submit their work. It also supports automated or manual grading, with results visible to students.

Results and Transcript

Generates and manages academic results, including grades, GPA calculations, and transcript generation for students.

Notification and Messaging

Provides alerts, updates, and internal messaging to keep students and staff informed of key activities, deadlines, and changes.

Admin Control Panel

Grants system administrators full control over system settings, user roles, academic calendar setup, reporting, and analytics.

Support and Feedback

Enables users to raise issues, ask questions, or submit feedback through a ticketing system, ensuring responsive technical and academic support.

3.14. VARIOUS MODULES OF THE METHOD

3.14.1. DATA DICTIONARY

User

Column	**Type**	**Null**	**Default**	**Comments**
_id	String	No		
firstName	String	Yes		
lastName	String	Yes		
email	String	No		
Password	String	No		
role	String	No		
googleId	String	Yes		
githubId	String	Yes		
profilePicture	String	Yes		
verifyOtp	String	Yes		

verifyOtpExpireAt	Number	Yes	0	
isVerified	Boolean	No	false	
resetOtp	String	Yes		
resetOtpExpireAt	Number	Yes	0	
Created_at	Date	No	Date.now()	

Admission

Column	**Type**	**Null**	**Default**	**Comments**
_id	String	No		
UserId	mongoose.Schema.Types.ObjectId	No		
firstName	String	No		
lastName	String	No		
nationality	String	No		
email	String	No		
countrycode	String	No		
Phone	String	No		
Address	String	No		
admissionType	String	No		
qualification	String	No		
Gender	String	No		
background	String	Yes		
dob	Date	No		
programType	String	No		
program	String	No		
referred	String	Yes		
referredEmail	String	Yes		
transcripts	Array of Strings	Yes		
testScore	Array of Strings	Yes		
Letter	Array of Strings	No		

idCardFront	Array of Strings	No		
idCardBack	Array of Strings	No		
statement	Array of Strings	No		
payment	Array of Strings	No		
health	Array of Strings	Yes		
extra	Array of Strings	Yes		
cv	Array of Strings	Yes		
status	String	No	pending	
applicationDate	Date	No	Date.now()	
created_at	Date	No	Date.now()	

Assignment

Column	Type	Null	Default	Comment
_id	String	No		
lecturerId	Mongoose.Schema.Types.ObjectId	No		
title	String	No		
course	String	No		
program	String	No		
attachedFiles	Array of strings	No		
description	String	No		
dueDate	String	No	Pending	
maxGrade	Number	No	100	
Created_at	Date	No	Date.now()	

assignmentSubmission

Column	Type	Null	Default	Comment
_id	String	No		
studentId	Mongoose.Schema.Types.ObjectId	No		
assignmentId	Mongoose.Schema.Types.ObjectId	No		

submittedFiles	Array of Strings	No		
Status	String	No	Pending	
Grade	Number	No	0	
Feedback	String	No		
Submitted_at	Date	No	Date.now()	

LiveClass

Column	**Type**	**Null**	**Default**	**Comment**
_id	String	No		
Host	Mongoose.Schema.Types.ObjectId	No		
Title	String	No		
Course	String	No		
Program	String	No		
Description	String	No		
startTime	Date	No		
endTime	Date	No		
Duration	Number	No	60	
Link	String	No		
recordedLink	String	No		
Passcode	String	No		
Attendees	Array of Mongoose.Schema.Types.ObjectId	No		
Status	String	No	scheduled	
Created_at	Date	No	Date.now()	

Notifications

Column	Type	Null	Default	Comment
_id	String	No		
recipientId	Mongoose.Schema.Types.ObjectId	No		
Message	String	No		
Read	Boolean	No	False	
Type	String	No	system	

3.14.2. RULES TO MOVE FROM ONE DATA MODEL TO ANOTHER

From Conceptual Model to Logical Model

- **Entities** are mapped to **collections**. For example, the Student, Lecturer, and Course entities become MongoDB collections.
- **Attributes** of each entity are mapped to **fields** inside documents. E.g., student.name, student.email, student.coursesEnrolled.
- **Relationships** are handled using either **embedding** (nested documents) or **referencing** (manual references):
    - **Embedding** is used when related data is accessed together (e.g., student profile and enrolled courses).
    - **Referencing** is used when related data is large or changes frequently (e.g., assignments linked to multiple courses).
- **Cardinalities** influence modeling:
    - **One-to-One**: Embed the sub-document directly.
    - **One-to-Many**: Embed if limited; reference if large or growing.
    - **Many-to-Many**: Use references or create a separate collection (e.g., registrations).

From Logical Model to Physical Model

- **Collections and Fields** are explicitly created based on logical structures.
- **Data types** are assigned using MongoDB-supported types; String, Number, Boolean, Date, Array, Object, ObjectId, etc.

- **Indexes** are added for performance on frequently queried fields (e.g., email, courseId, studentId).
- **Validation rules** (optional) can be added using **schema validation** with Mongoose or MongoDB's built-in validators to enforce field types and required fields.
- **Relationships** are managed using manual references (storing ObjectIds of related documents) or through embedding.

3.15. SOFTWARE USED

Visual Studio Code

This was the IDE used for building this web application.

MongoDB Compass and MongoDB Server

MongoDB server was a database server on which my database ran on, and MongoDB compass was the software used to interact visually with the database.

Postman

Postman was used to test my API Endpoints written with Node.js and express to see if they're actually working or not, before consuming them on the frontend.

E-Drawmax

This is software was used in the drawing the various logical diagrams such as UML Class diagram, Use case diagram, and Sequence diagram.

YouTube

Whenever I got lost, I watched tutorial videos on YouTube which helped me find myself and get past any obstacles I'd faced.

3.16. PROGRAMMING LANGUAGES USED

NEXT.JS (With TypeScript)

Next.js is a modern web development framework built on top of **React**, maintained by Vercel, and optimized for building fast, scalable, and user-friendly web applications. It is used to develop server-rendered or statically generated websites and web apps with excellent performance, SEO support, and developer experience.

NODE.JS (With TypeScript)

Node.js is a JavaScript runtime environment built on Chrome's V8 engine that allows developers to run JavaScript code outside the browser, particularly on the server side. It is designed for building scalable network applications and is widely known for its event-driven, non-blocking I/O model, which makes it lightweight, efficient, and suitable for real-time applications.

Node.js is commonly used to build web servers, RESTful APIs, microservices, and real-time applications such as chat systems and collaborative tools. It has a vast ecosystem of open-source libraries available through npm (Node Package Manager), making it easy to integrate a wide range of functionalities quickly.

3.17. HARDWARE USED

PC (Laptop)

A computer with windows 10 operating system was used in implementing the system. The computer has a Hard Disk Space of 250 Gigabytes SSD, a RAM of 8 Gigabyte, a 64-bits Operating System and Intel core i7 4CPUs with of speed 2.5GigaHertz extensible up to 3.5GHz.

Orange’s Infinity Box WIFI

Monthly internet subscriptions for unlimited internet were made to Orange in exchange for a 10Mbps (10 Megabits per second) internet speed.

3.18. MODULES OF THE DESIGNED SYSTEM

The system was developed with the help of the following modules:

- A MongoDB database, hosted on a virtual private server, to store and manage all user, course, and academic data.

A web-based application that interacts with the database and presents the following modules:

Common Landing Page

A homepage accessible to all visitors with options to log in, apply for admission, or learn more about the university.

User Login Module

A secure module that allows students, lecturers, and admins to log into the system using their credentials.

Password Recovery Module

Provides users with the ability to reset forgotten passwords via email verification.

Admission Application Module

Allows prospective students to fill out and submit admission forms online with required documents.

Content Management Module

Allows lecturers to upload lecture materials, notes, assignments, and multimedia resources.

Assignment & Exam Module

Supports uploading of assignments by lecturers and submission by students, as well as taking online quizzes/exams.

Live Class Module

Connects students and lecturers in real-time for online lectures using embedded video conferencing tools.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1. INTRODUCTION

This chapter talks about the Virtual University Platform's design, looks, and how each component was placed to ensure user friendliness, and mobile-responsiveness.

4.2. RESULTS

We shall display print screens of the various modules and give their functionalities, in order to better understand the relationship between the lecturers, students, and system administrators.

4.3. PRESENTATION OF SCENARIOS

Since this is a web application that allows students to be able to go through the University, some security measures may be applied within the system to keep it protected and less vibrant to attackers. This system will only direct the various users to their various profiles and dashboards, depending on the nature of their account.

LANDING PAGE

These are the first pages that gets displayed when a user visits the web application. Users are not required to be logged in before being able to view the landing page. The landing page simply describes what the app is all about, and gives information about interested students and lecturers.

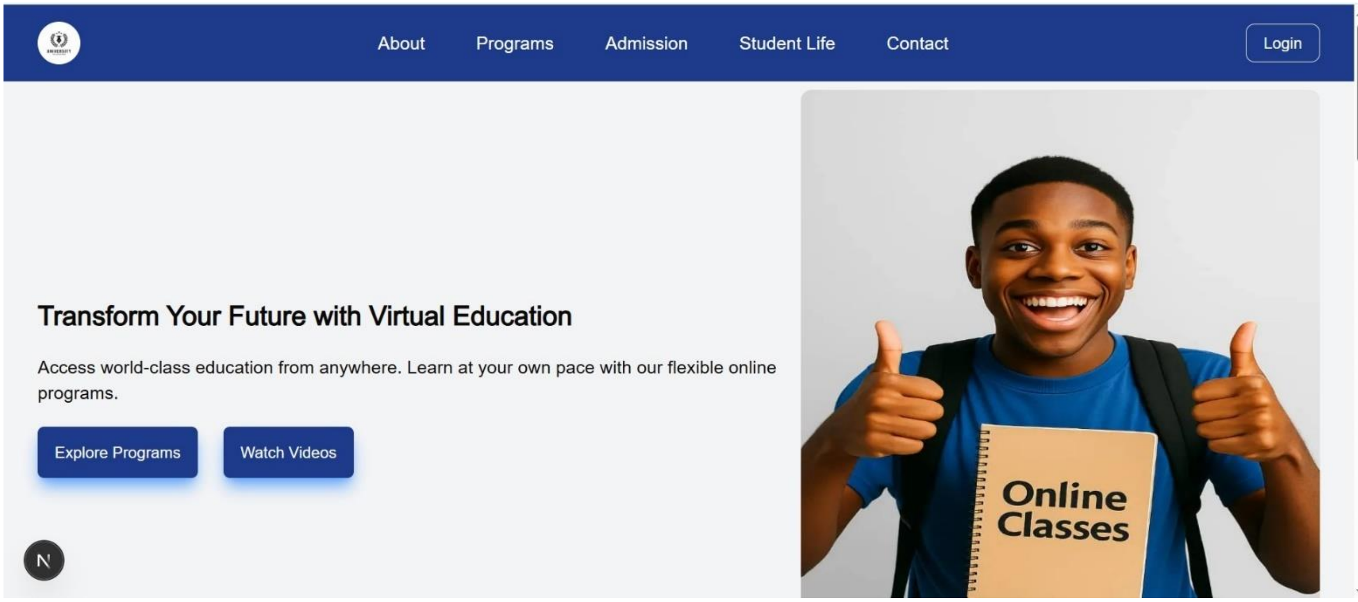

Figure 5: Landing page without log in

LOGIN PAGE

This page enables users to be able to login into the system if they have an account, or click on the signup link to create one. This page supports the ability for users to login using their google account, GitHub account, or through the normal email and password. Users are also able to recover their passwords when they click on forgot password through OTP messages sent to their email addresses.

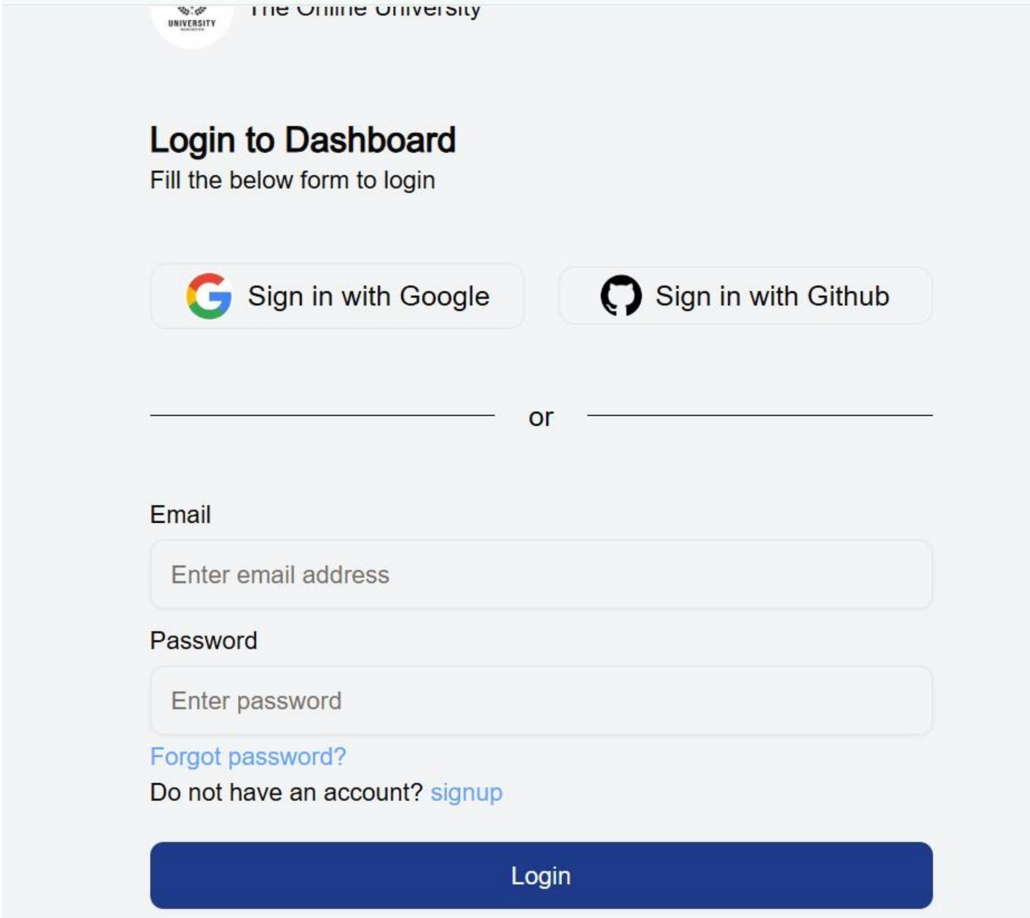

Figure 6: Login page

LANDING PAGE AFTER LOGIN

This is the landing page after login. I am currently logged in as an administrator. Users can verify their account using their email if they didn't on creating the account. They are given a grace period of 2 days before their account gets disabled if they don't verify it using their email address.

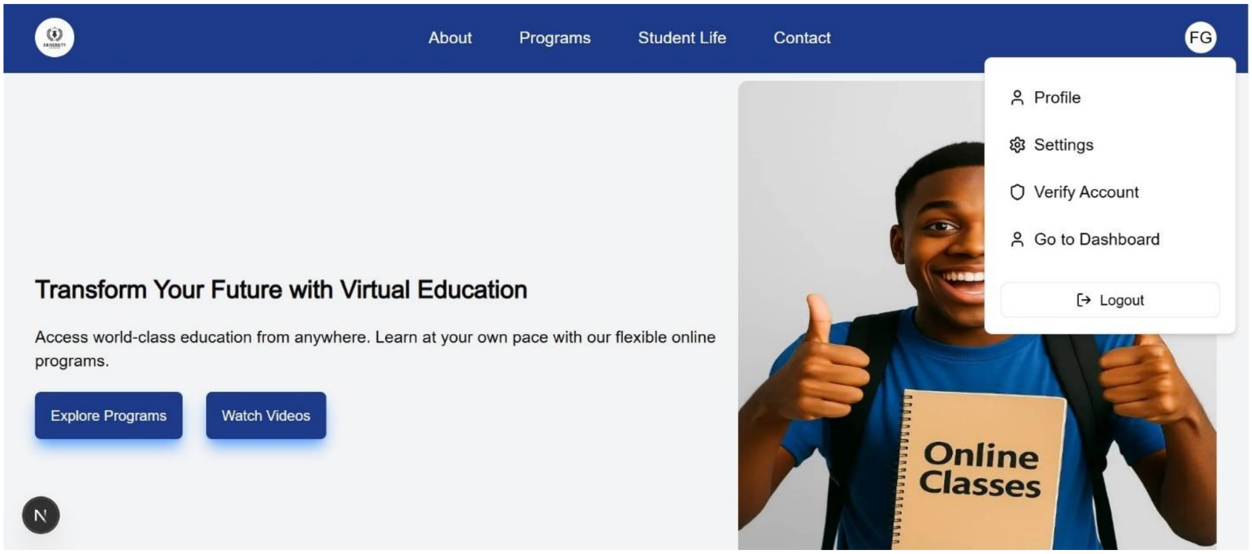

Figure 7: Landing page after login

ADMISSION PAGE

This is the admission page, and it allows users to be able to submit an application to get into the university. It is in three steps. The first step asks for basic information, while the second asks for important documents necessary for you to gain admission either as student or as lecturer, and the last stage is for you to upload proof of application fee and other helper documents.

The user’s role on this system depends on their application type. You have the ability to submit your application either as student or as admin, and the document required for each of them will be asked of you accordingly. Once you submit as a student, your role becomes student, and same for lecturer. Also, every user can only submit only 1 application at a time and will have to wait for it to either get accepted or rejected. They can submit another if their initial one gets rejected. They can also decide to delete the current application and resubmit a new one.

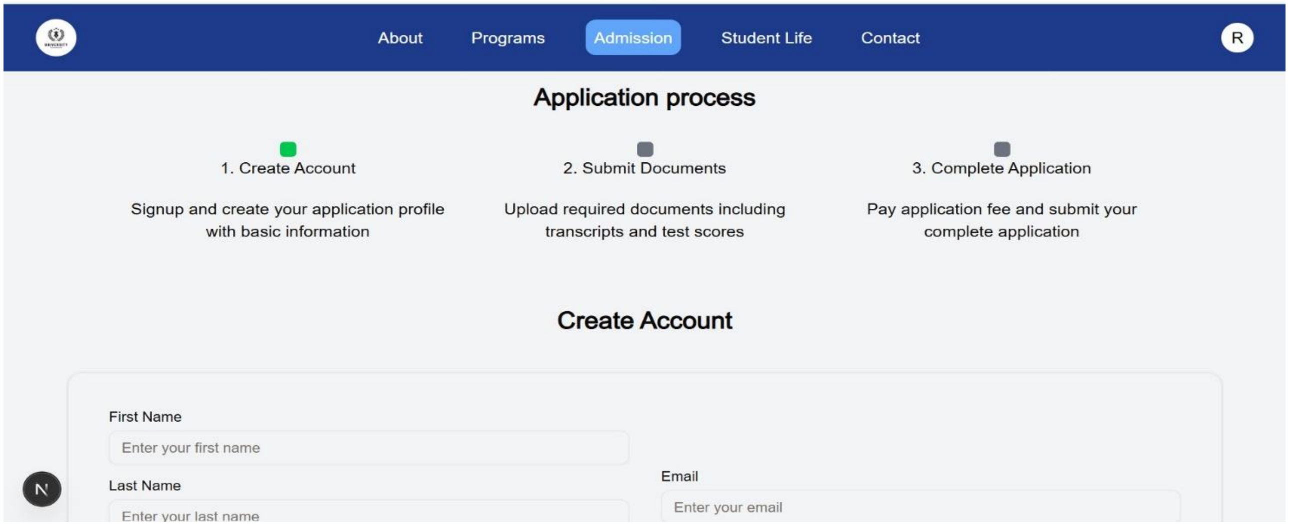

Figure 8: admissions page

VIEW ALL APPLICATIONS

Here, the admin can view all applications sent by the users and can either accept, or reject a user's application. Once they admin accepts the application, the student or lecturer automatically gets admitted into the university and can now go to the students' or lecturers' dashboard. The admin can click on the right arrow for every application to see all information about the application, view the applicant's files, and decide to either accept or reject them.

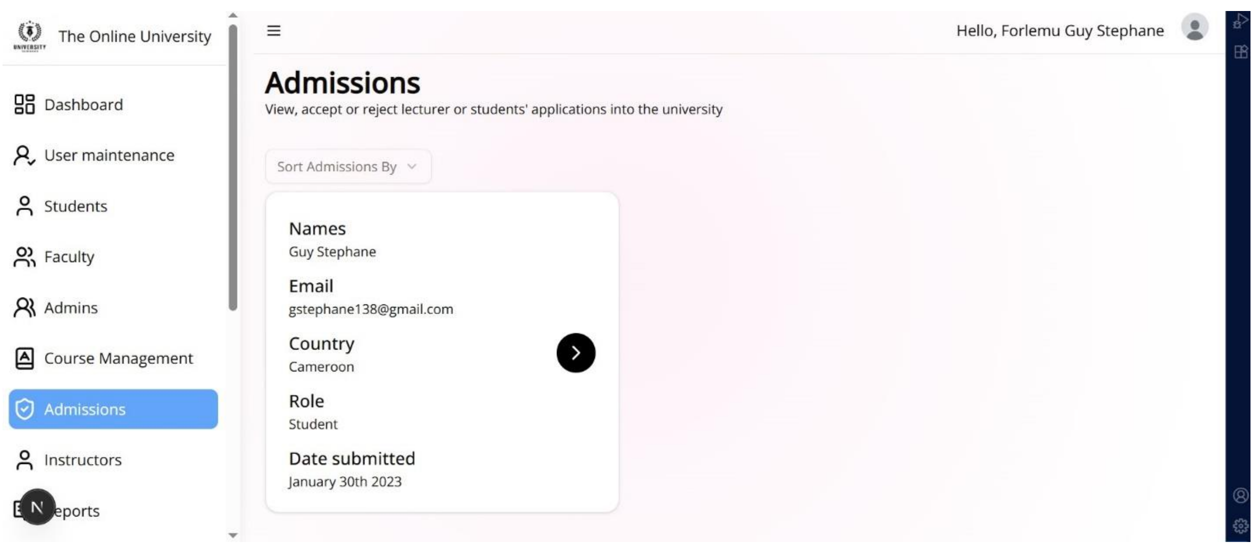

Figure 9: Admissions page

LECTURER DASHBOARD

This is the lecturer's dashboard, which the lecturer can start seeing after their application has been accepted.

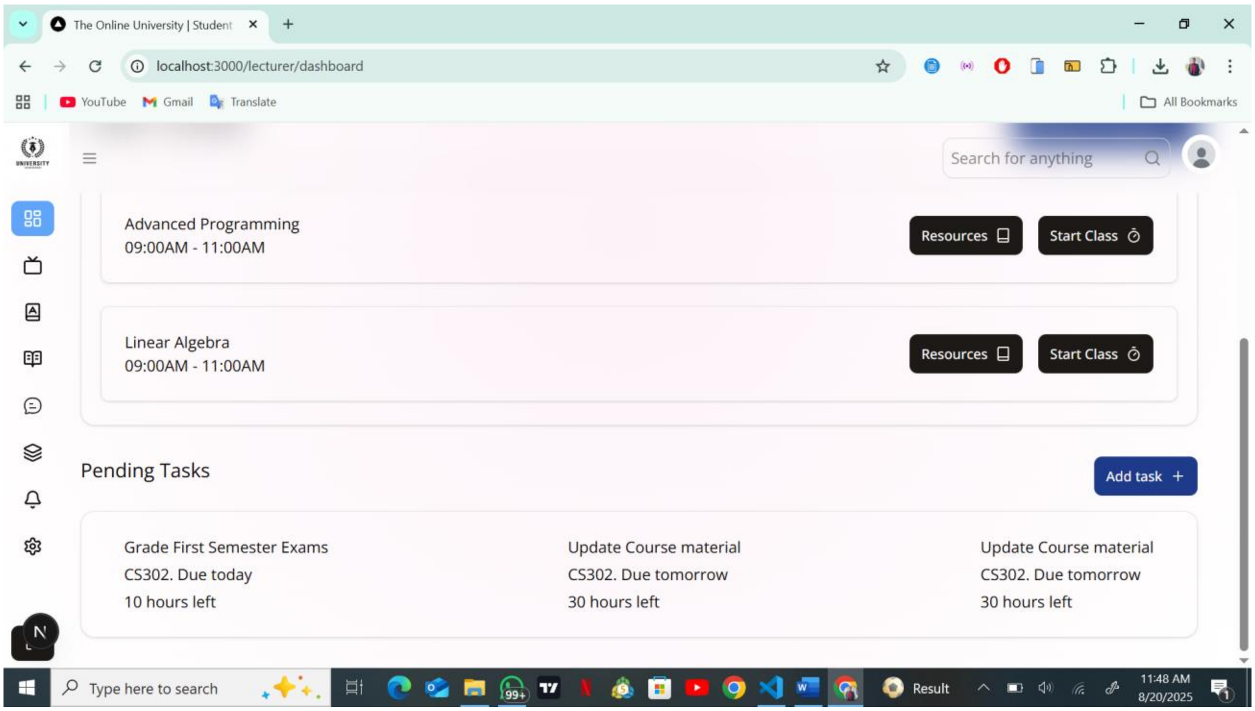

Figure 10: Lecturer's dashboard

CLASS PAGE

This page gives the lecturer the ability to start an instant class, join an existing one, or to be able to create a scheduled class.

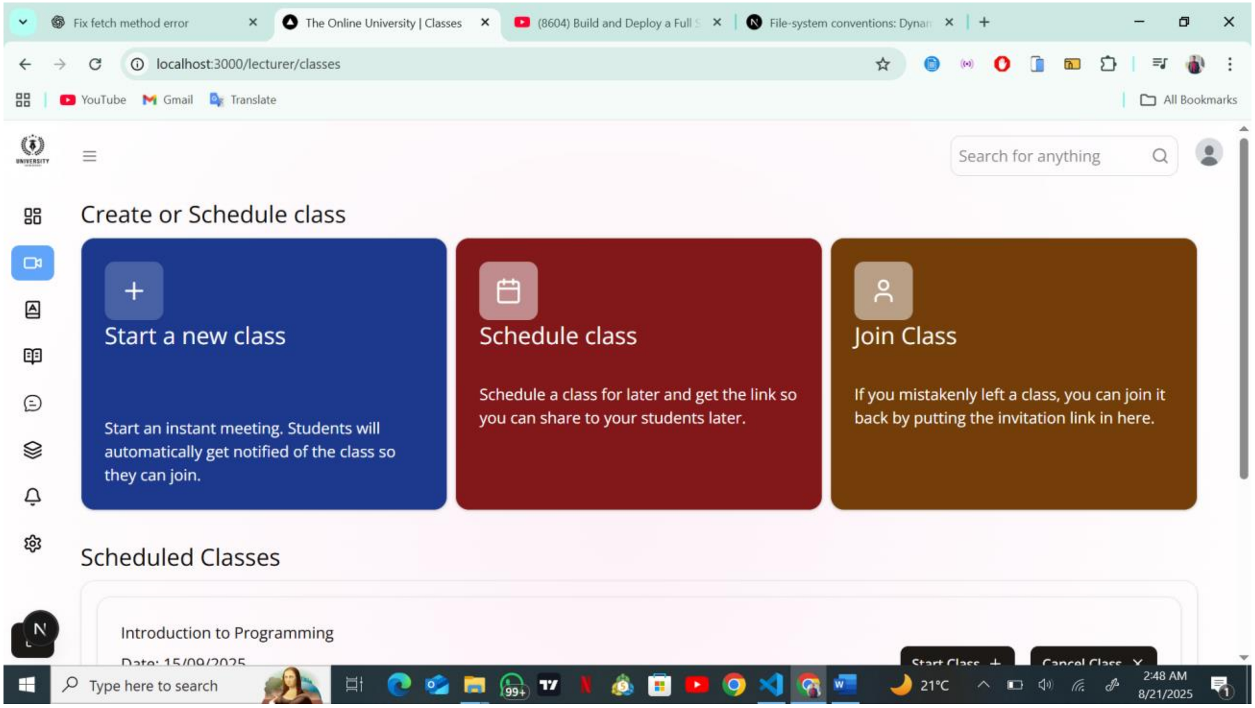

Figure 11: Lecturer classes page

SCHEDULE CLASS

The lecturer can fill this form to schedule the class.

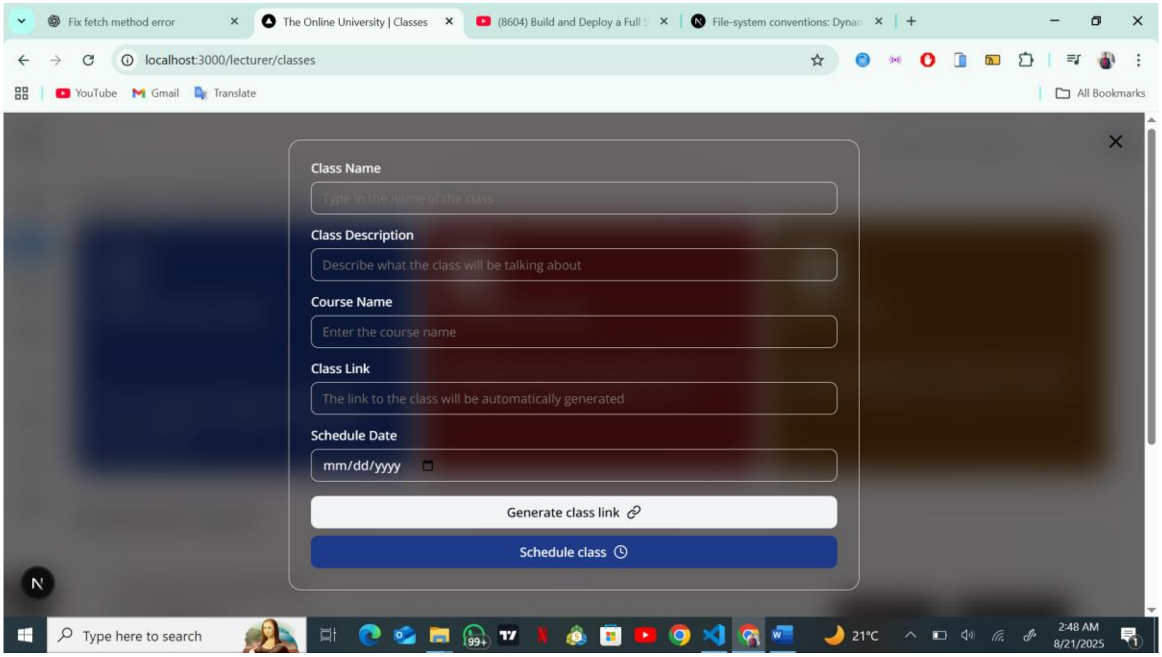

Figure 12: Lecturer schedule class page

CALL SETUP PAGE

Both the lecturer who starts the class and the attendees (students) get redirected to the call setup page where they can setup their mic and camera before joining the class

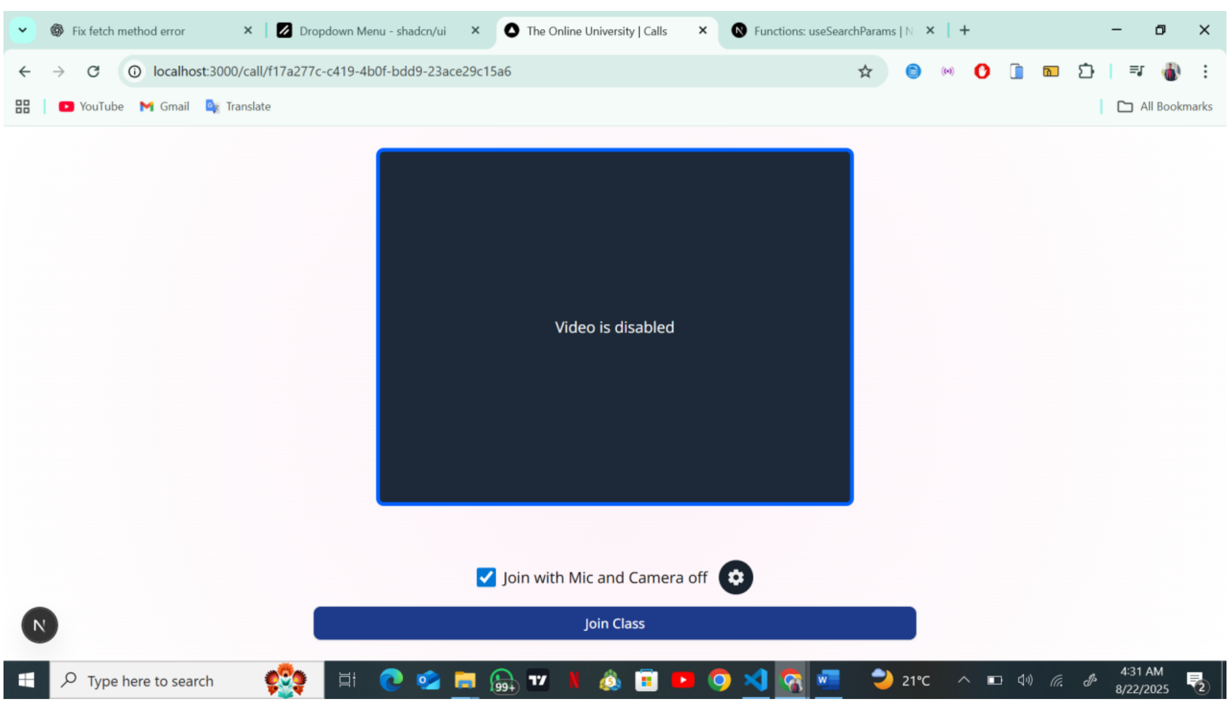

Figure 13: Class setup

LIVE CLASS PAGE

As you can see below, we currently have two users who've joined the call. GS is the lecturer and RB is a student. Both lecturer and student have several control options below, including the ability to share screen. The creator of the call can end the call for everyone as well.

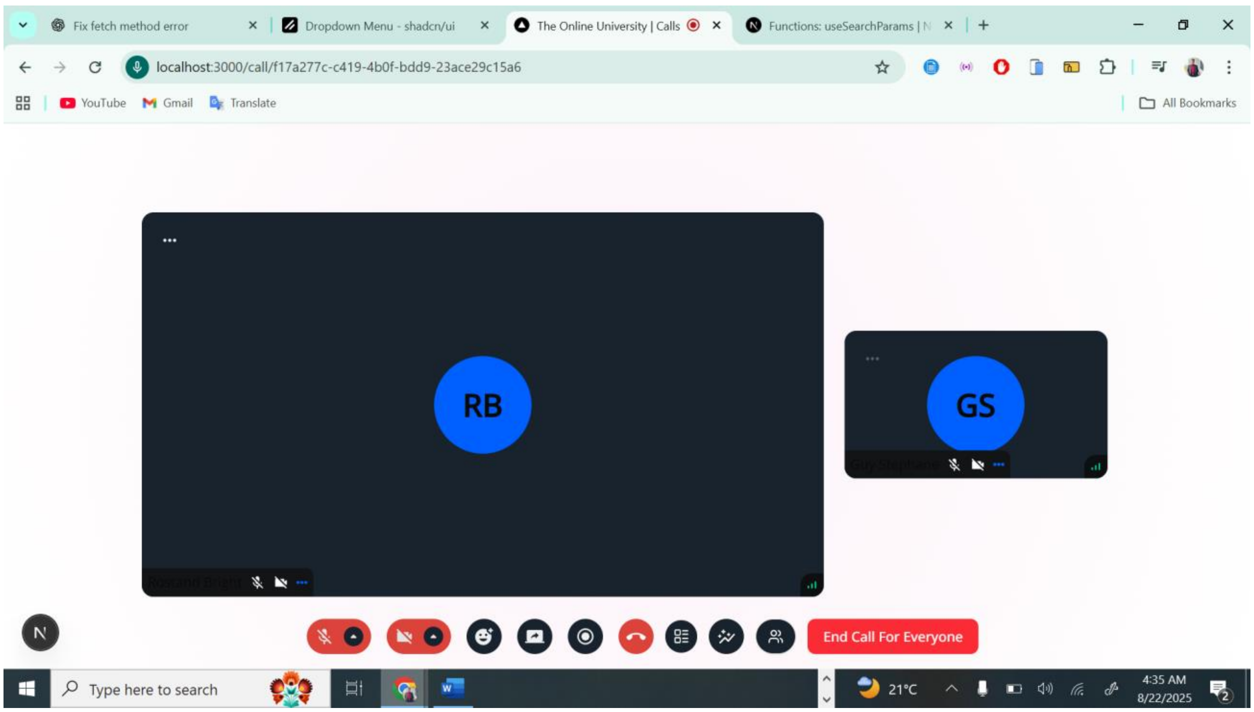

Figure 14: Live class

ASSIGNMENT PAGE

This page allows lecturers to be able to upload assignments to the system so that the students who are currently enrolled in that course can upload files in response to the assignments uploaded by their course lecturer.

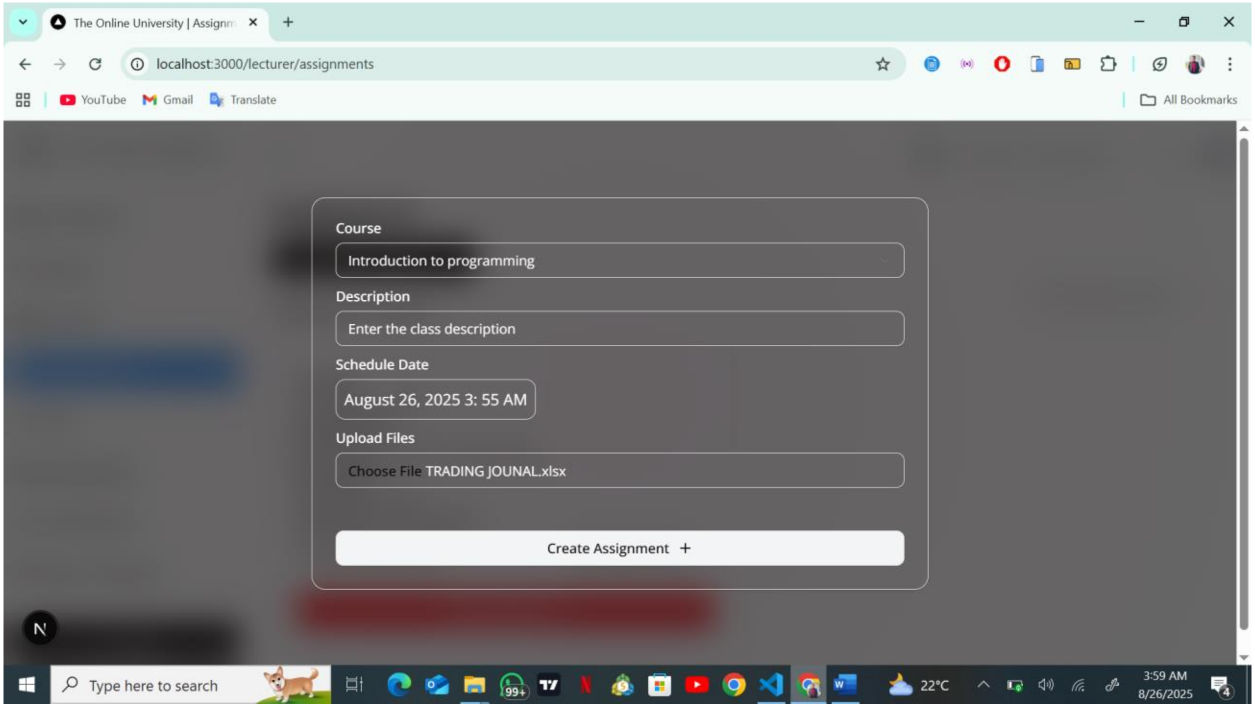

Figure 15: add assignment modal

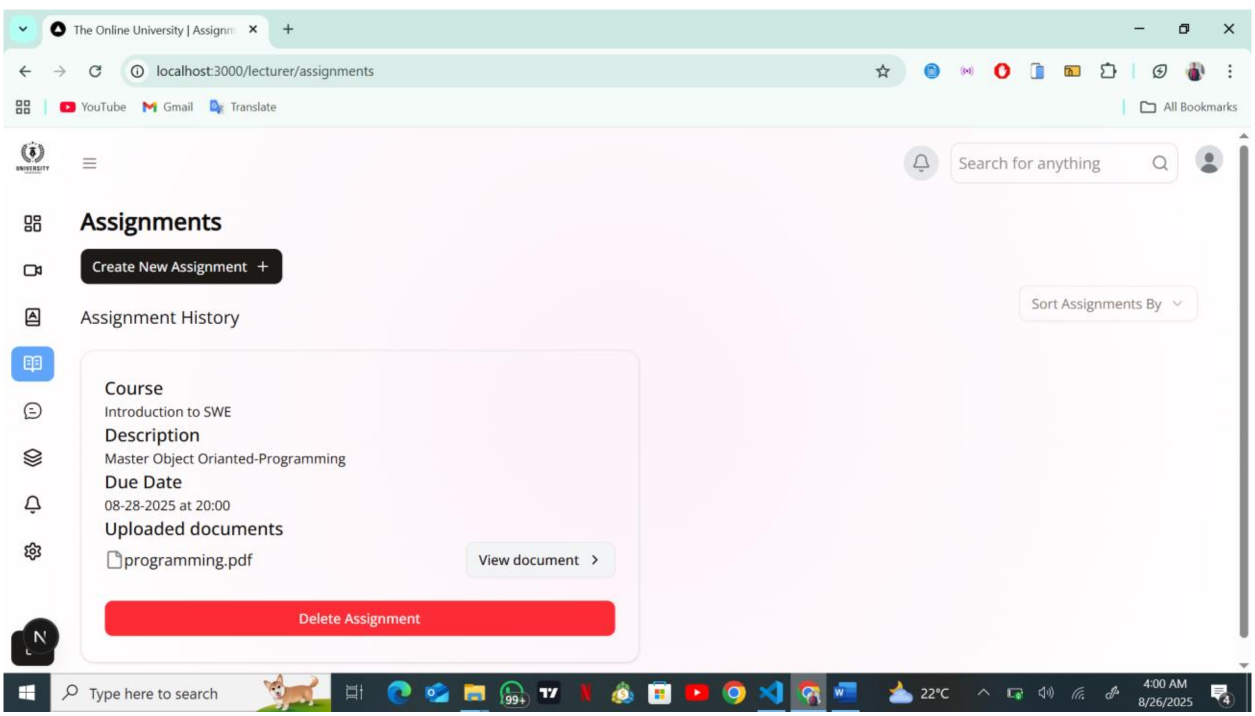

Figure 16: The assignments home page

CHAPTER FIVE

CONCLUSIONS AND RECOMMENDATIONS SUMMMARY OF FINDINGS

5.1. SUMMARY OF FINDINGS

The end of this work marks the beginning of many achievements which have been accomplished according to the objectives formulated. First and foremost, this platform has been designed using the 3-tier architecture which is consist of the server, client and the database all interacting with each other through Node.js at the level of the control structure to ensure the security of the personnel information to avoid hacking of the system or any other undesired intrusion. The implementation of this platform was a challenging one as it requires the knowledge of many programming languages and tools, to make it function as to meet the user's needs and standards.

5.2. DIFFICULTIES ENCOUNTERED

Poor internet connection

This made research to be slow, and changes on the mobile app to be implemented very slowly.

Difficulty in designing the platform

Building some of the APIs at the level of the backend, such as the API that handles file uploads and submission of data to the data base on the admissions page was quite challenging.

5.3. RECOMMENDATIONS

The benefits of using such a system in the academic process are substantial. An online university management system can help students, instructors, and administrators organize and manage academic activities in a centralized, easily accessible platform. This is especially valuable for distance learning environments, where efficient communication, streamlined processes, and digital access to resources are essential for success.

By using an online university management system, students can conveniently register for courses, access lecture materials, submit assignments, attend live classes, and track their academic progress from anywhere. This flexibility enhances learning opportunities and makes education more inclusive, especially for individuals balancing work, family, and studies.

Additionally, the system improves efficiency for instructors and administrative staff by automating routine processes such as grading, notifications, scheduling, and admissions. This not only saves time but also minimizes the risk of errors in student records and academic documentation.

The system also supports better student engagement. With integrated live class features, discussion forums, and notification systems, learners can interact with instructors and peers in real time, fostering a sense of community despite the remote setting.

Furthermore, an online university system serves as a valuable tool for academic planning. Students can track their course requirements, monitor grades, and receive timely reminders about deadlines, while administrators can access real-time data to make informed institutional decisions.

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APPENDIX

Showing my visual studio code setup and files.

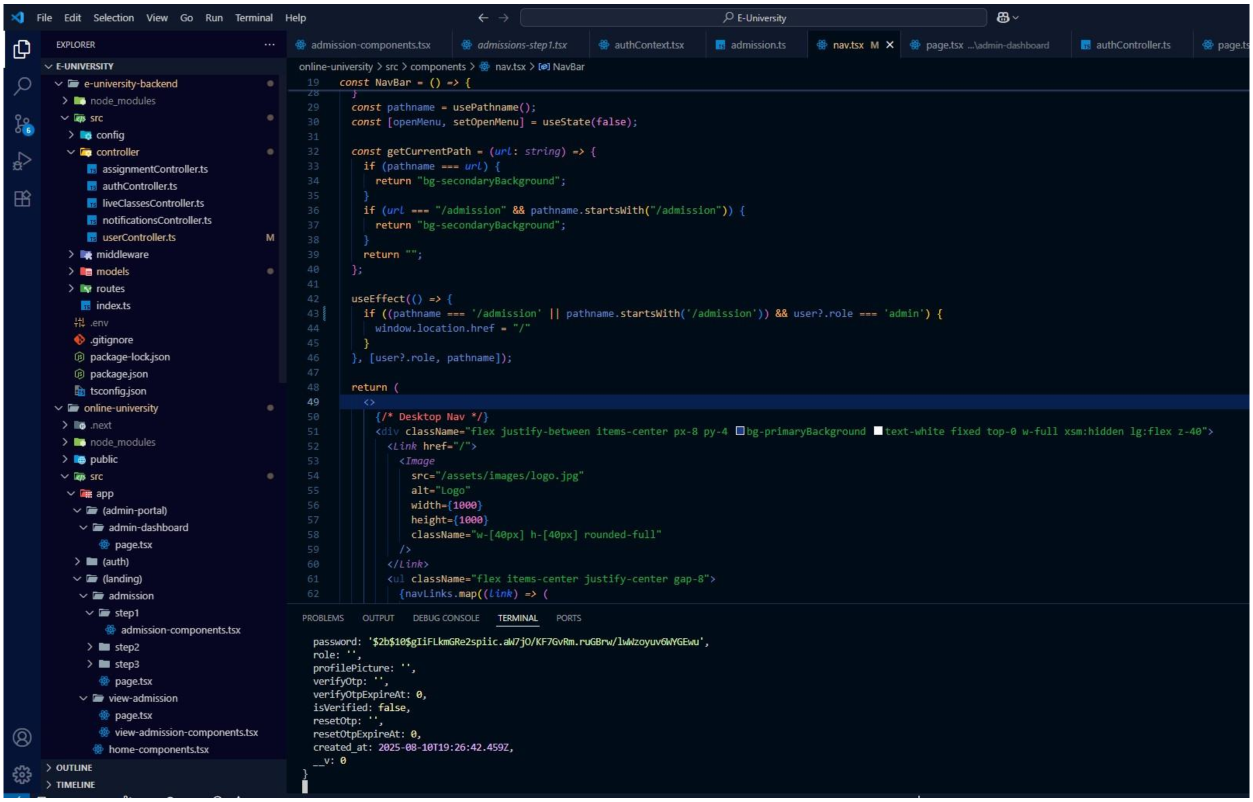

Figure 17: Display of codebase

Code that creates the admission schema

e-university-backend > src > models > Admission.ts > admissionSchema

```
import mongoose from "mongoose";

const admissionSchema = new mongoose.Schema({
  userId: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true },

  firstname: { type: String, required: true },
  lastname: { type: String, required: true },
  nationality: { type: String, required: true },
  email: { type: String, required: true },
  countrycode: { type: String, required: true },
  phone: { type: String, required: true },
  address: { type: String, required: true },

  admissionType: {
    type: String,
    required: true,
    enum: ["student", "lecturer"],
  },

  qualification: { type: String, required: true },
  gender: { type: String, required: true, enum: ["male", "female"] },
  background: { type: String },
  dob: { type: Date, required: true },

  programType: {
    type: String,
    required: true,
    enum: ["undergraduate", "postgraduate"],
  },
  program: { type: String, required: true },

  referred: { type: String, default: "" },
  referralEmail: { type: String, default: "" },

  // File/document uploads
  transcripts: [{ type: String }],
  testScore: [{ type: String }],
  letter: [{ type: String }],
  idCardFront: [{ type: String, required: true }],
  idCardBack: [{ type: String, required: true }],
  statement: [{ type: String, required: true }],
  payment: [{ type: String, default: "" }],
  health: [{ type: String, default: "" }],
  extra: [{ type: String, default: "" }],
  cv: [{ type: String, default: "" }],

  documents: [{ type: String }],

```

Figure 18: Admission schema

API that submits a student or lecturer's admission application to the database.

```
export const submitAdmissionApplication = async (
  req: Request,
  res: Response
) => {
  const { userId, applicationData } = req.body;
  try {
    const user = await User.findById(userId);

    if (!user) {
      return res.status(404).json({
        success: false,
        message: "Please login before submitting your application",
      });
    }

    const existingAdmission = await Admission.findOne({
      $or: [{ userId }, { email: user.email }],
    });
    if (existingAdmission) {
      return res.status(400).json({
        success: false,
        message: "You have already submitted an application",
      });
    }

    if (user.email !== applicationData.email) {
      return res.status(400).json({
        success: false,
        message: "Email does not match with the logged-in user",
      });
    }

    user.firstName = applicationData.firstname;
    user.lastName = applicationData.lastname;
    user.role = applicationData.admissionType;
    await user.save();

    const admission = new Admission({
      userId: user._id,
      ...applicationData,
    });

    await admission.save();
    return res.status(201).json({
      success: true,
```

Figure 19: Api for admission application submission

My authContext file setup for state management and ensuring user auth and user sessions, etc.

```
online-university > src > context > authContext.tsx > AuthProvider > logout
"use client";

import { API_URL } from "@/API";
import {
  createContext,
  useContext,
  useState,
  ReactNode,
  useEffect,
} from "react";
import { toast } from "sonner";
import { admissionFormData } from "@/types/admission";

interface User {
  firstName: string;
  lastName: string;
  profilePicture: string;
  email: string;
  role: string;
  isVerified: boolean;
  otp?: string;
}

interface AuthContextType {
  user: User | undefined;
  admissionData: admissionFormData | undefined;
  isAuthenticated?: boolean;
  admissionLoading?: boolean;
  fetchUser: () => Promise<void>;
  logout: () => Promise<void>;
  loading?: boolean;
  checkUserAuthentication: () => Promise<boolean>;
  setUser: React.Dispatch<React.SetStateAction<User | undefined>>;
}

const AuthContext = createContext<AuthContextType | undefined>(undefined);

export const AuthProvider: React.FC<{ children: ReactNode }> = ({
  children,
}) => {
  const [user, setUser] = useState<User>();
  const [admissionData, setAdmissionData] = useState<admissionFormData>();
  const [loading, setLoading] = useState<boolean>(true);
  const [admissionLoading, setAdmissionLoading] = useState<boolean>(true);
  const [isAuthenticated, setIsAuthenticated] = useState<boolean>(false);

  // Fetch user data from the backend
  const fetchUser = async () => {
```

Figure 20: Auth context file showing auth setup